

Bandgap Reference Verification

Rapid Adoption Kit (RAK)

Product Version: IC6.1.8 ISR21, Spectre20.1 ISR10
December 2021

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Purpose

This RAK introduces Bandgap Reference Voltage supply variation across PVT corners, PSRR, Stability, and Transient Noise setup examples. The document helps understand the usage of dc/dc with temperature sweep, ac, stb, and transient noise analysis to measure the various Bandgap design specifications based on different testbench setups.

Audience

This RAK is intended for ADE Explorer and Spectre users who want to measure Bandgap variations, Stability, PSRR, and Transient Noise of a Bandgap.

Tool Versions

This RAK has been verified using IC 6.1.8 ISR21 and Spectre20.1 ISR10 with Spectre/SpectreX Mode.

Overview

1.1 Design Example

Bandgap references (BGR) are fundamental circuits, which are used to provide reference supply that has little dependence on temperature (temperature independent).

You can generate a temperature-independent quantity (Voltage and Current) by summation of Negative-temperature coefficient quantity and Positive-temperature coefficient quantity.

It has been implemented using the Cadence generic PDK 45nm CMOS process.

The key analog components of the BGR are the Positive-temperature coefficient and Negative-temperature coefficient generation with BJT used as diode, OpAmp, and startup circuit.

1.2 Lab Overview

BGRs are fundamental circuits that are used to provide reference supply (temperature independent) in integrated circuits.

The Bandgap circuit used in this RAK is based on the voltage summing topology.

- Temperature sweep analysis determines temperature variations on the output reference supply.

- Loop gain and phase determine whether a negative-feedback loop of a BGR is stable.
- PSRR is a measure of the influence of power supply ripple on the BGR output voltage.
- Generally, several devices contribute towards the noise in a BGR circuit. Transient noise calculates the effects of large signals on virtually any system. It also determines the impact of noise in the time domain.

This RAK consists of examples on the following topics:

- Measurement of temperature variation using dc sweep analysis
- PSRR of a BGR using ac analysis
- Loop gain and phase margin of a BGR using stability analysis
- Large signal noise of a BGR using transient noise analysis

Setting Up the RAK Database

Action 1: Unzip and untar the TB_BGR.tar.gz database.

```
Unix> tar -zxvf TB_BGR.tar.gz
```

Action 2: Navigate to the TB_BGR directory.

```
Unix> cd TB_BGR
```

Lab 1: Output Variation Measurements Using DC Analysis

- DC analysis (temperature sweep) is used to measure the BGR output variation and accuracy ($V/^{\circ}C$).
- DC analysis finds the DC operating-point or DC transfer curves of the circuit. To generate transfer curves, specify a parameter and a sweep range.
- Circuit temperature is used as a swept parameter.

Action 1: Invoke **Virtuoso**.

```
Unix> virtuoso &
```

Action 2: Open the Library Manager from the CIW window by navigating to **Tools > Library Manager**.

Action 3: From the Library Manager, open the **TB_BGR > tb_bgr > schematic** view.

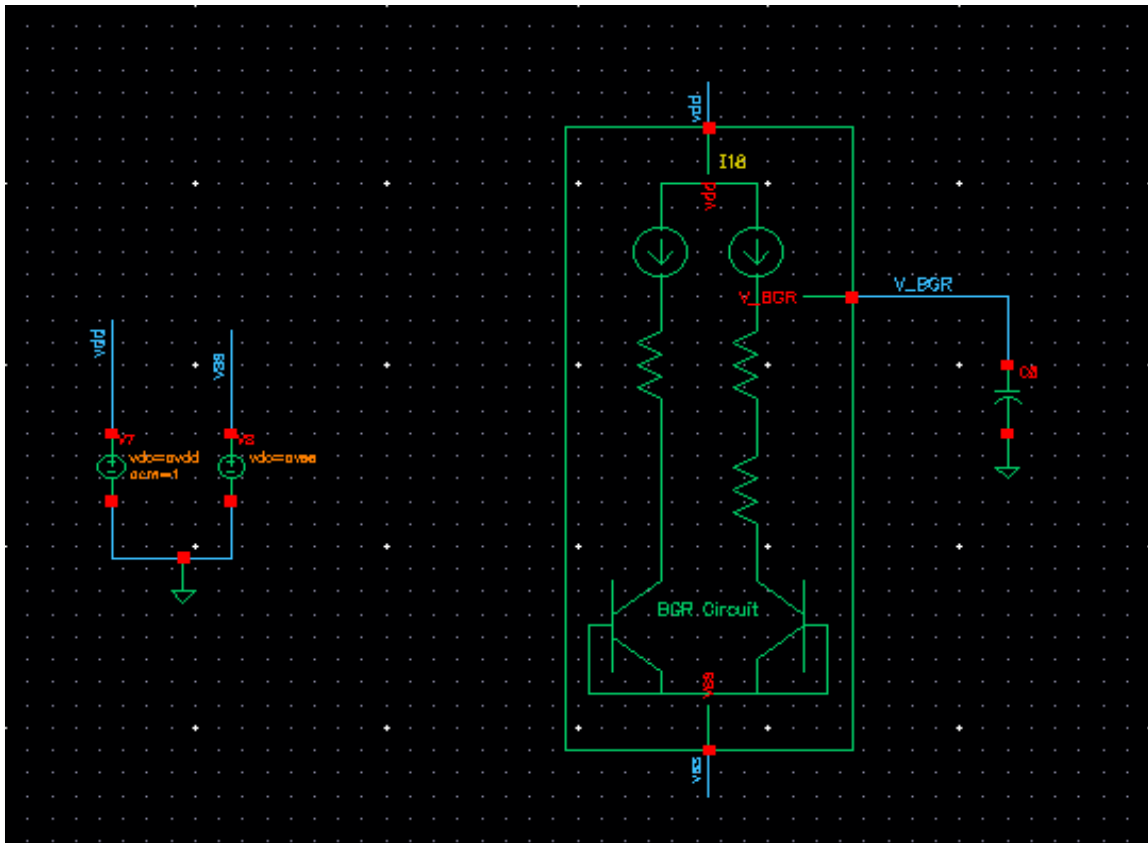


Figure 1: Bandgap testbench schematic

Action 4: From the schematic, select **Launch > ADE Explorer**.

Action 5: In the **Launch ADE Explorer** window, select **Create New View** and click **OK**.

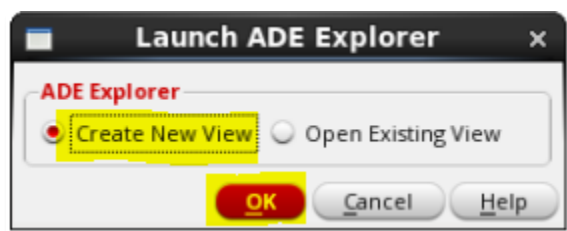


Figure 2: Launching ADE Explorer from schematic window

Action 6: In the **Create new ADE Explorer view** window, select **Library > TB_BGR**, **Cell > tb_bgr**. Type the view as `maestro_Dc` and click **OK**.

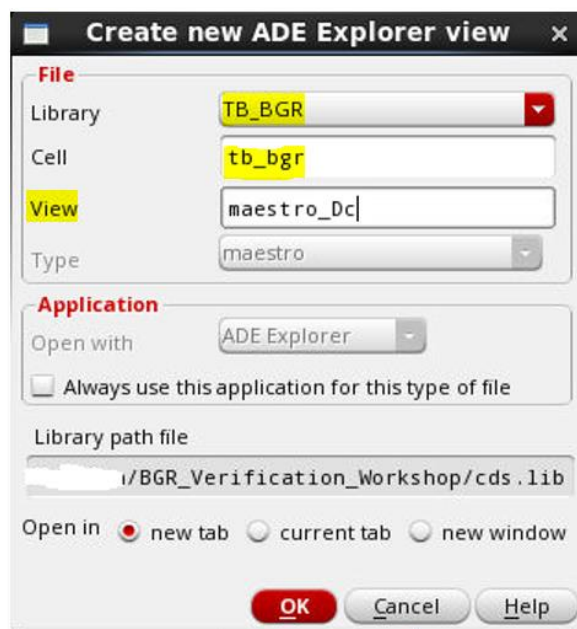


Figure 3: Opening maestro_Dc view of tb_bgr cell

The **ADE Explorer** window opens as shown in Figure 4.

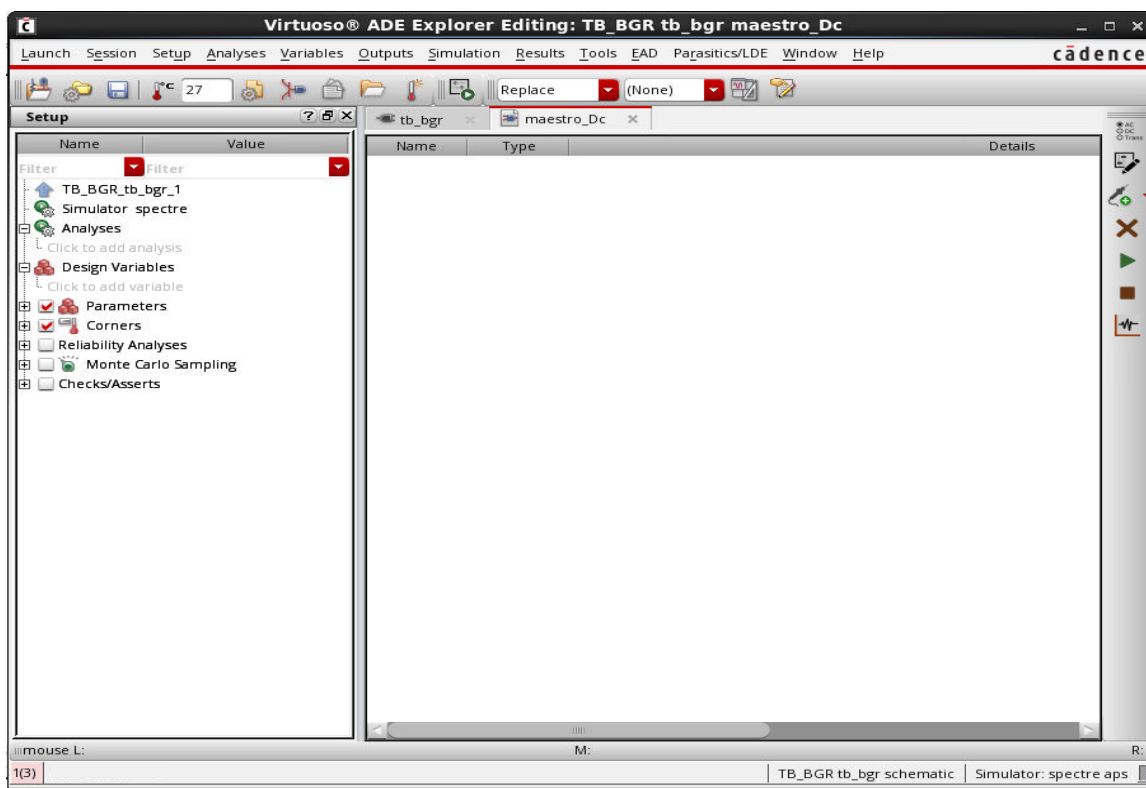


Figure 4: ADE Explorer window showing maestro_Dc view of tb_bgr cell

Action 7: In the **ADE Explorer** window, select **Setup > Model Libraries** as shown in Figure 5.

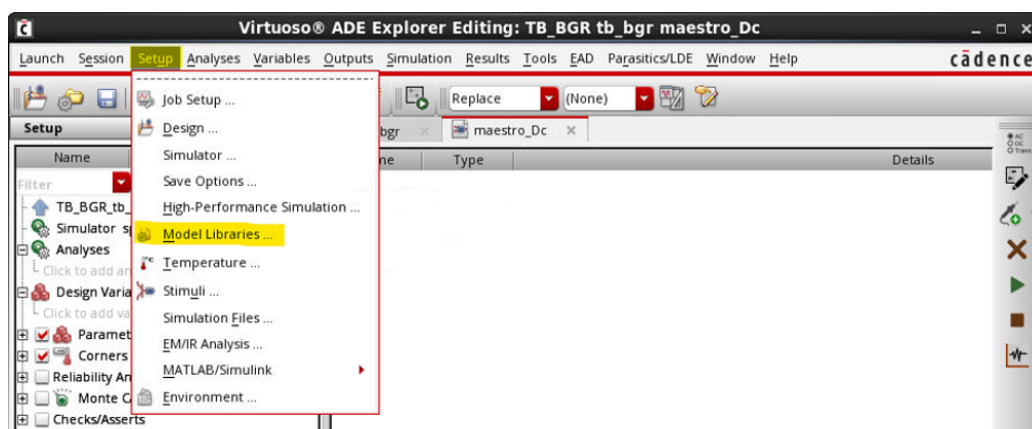


Figure 5: Opening Model Library Setup form from ADE Explorer window

The **Model Library Setup** form opens as shown in Figure 6.

Action 8: Browse the **gpdK045.scs** model file in the **Model File** section.

Action 9: Set the **Section** to **tt** in the **Model Library Setup** form. Then, click **OK**.

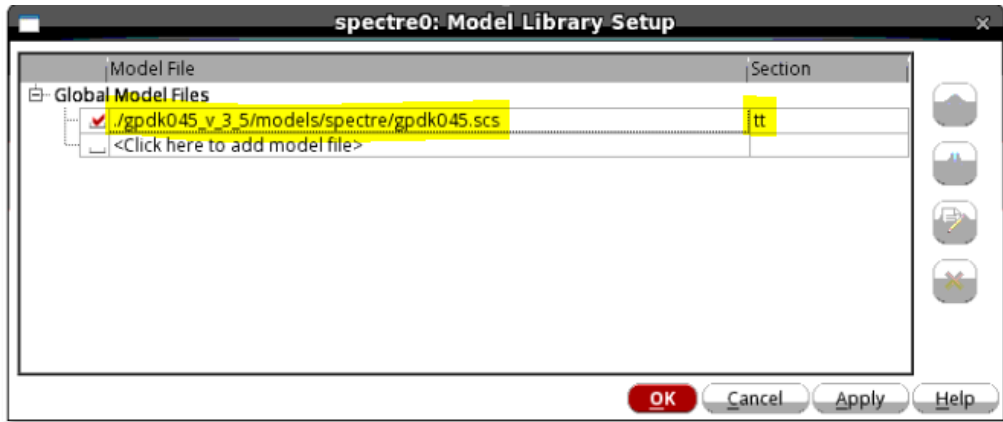


Figure 6: Set model file in Model Library Setup form

Action 10: In the ADE Explorer **Setup** assistant, select **Click to add analysis** in the **Analyses** section of the **Setup** assistant.

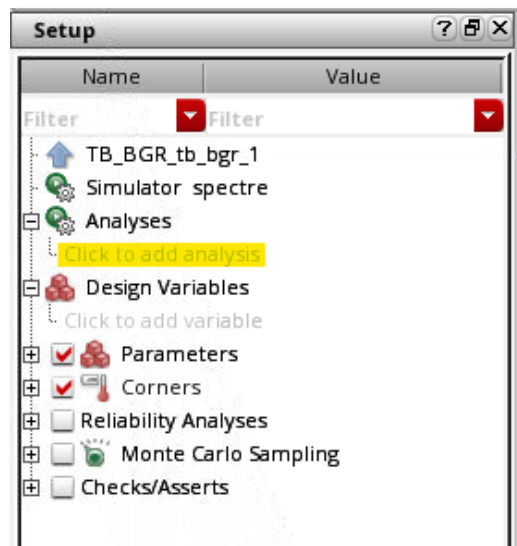


Figure 7: Opening 'Analyses' form from 'Click to add analysis'

Action 11: In the **Choosing Analyses** form, select the **dc** analysis with the following setup. Then, click **OK**:

- Enable **Save DC Operating Point**.
- Select **Sweep Variable** as **Temperature**.
- For **Sweep Range**, set **Start** to **-40** and **Stop** to **120**.
- Choose **Linear** for **Sweep Type** with step size of **10**.

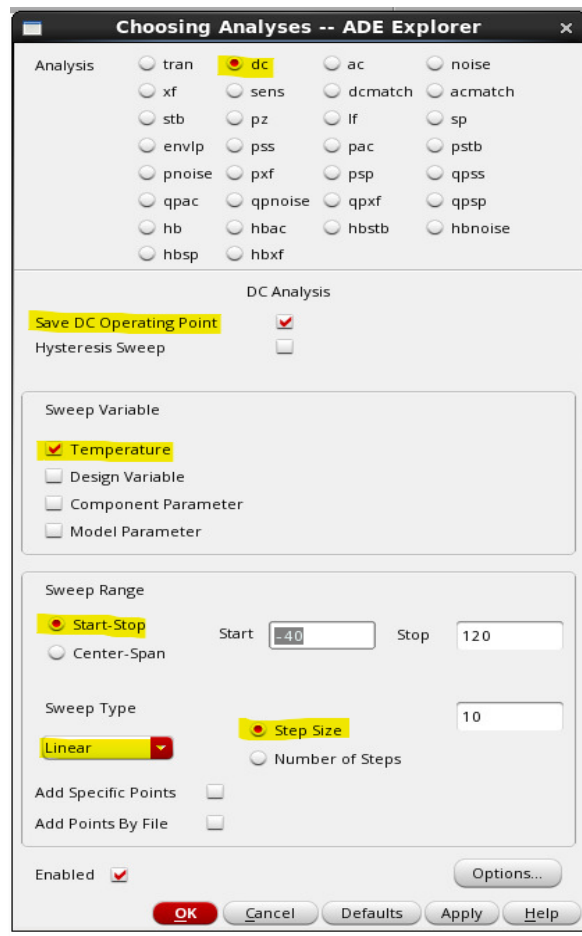


Figure 8: Setup of DC-Sweep Analysis

Action 12: In the **ADE Explorer** window, select **Outputs > Save All** as shown in Figure 9.

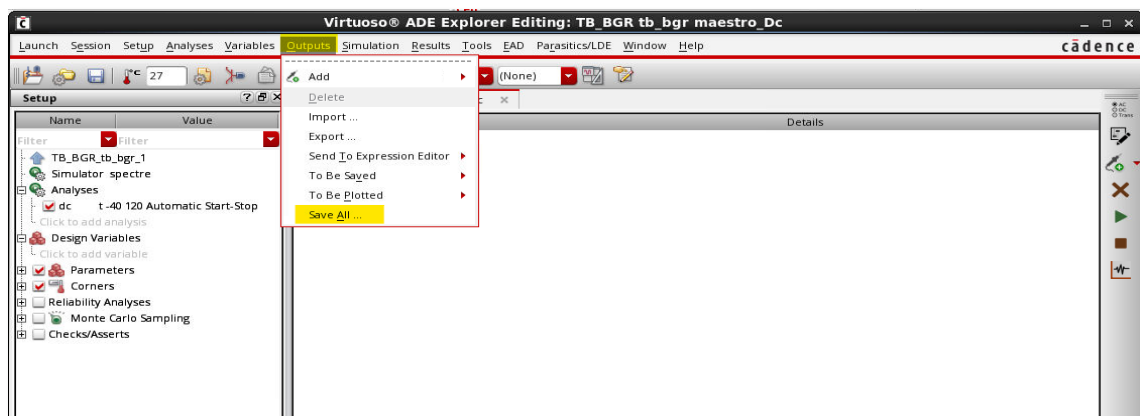


Figure 9: Options for saving outputs

Action 13: Enable **all** for **Select device currents**. For **Select signals to output**, **allpub** will be enabled by default. Then, click **OK**.

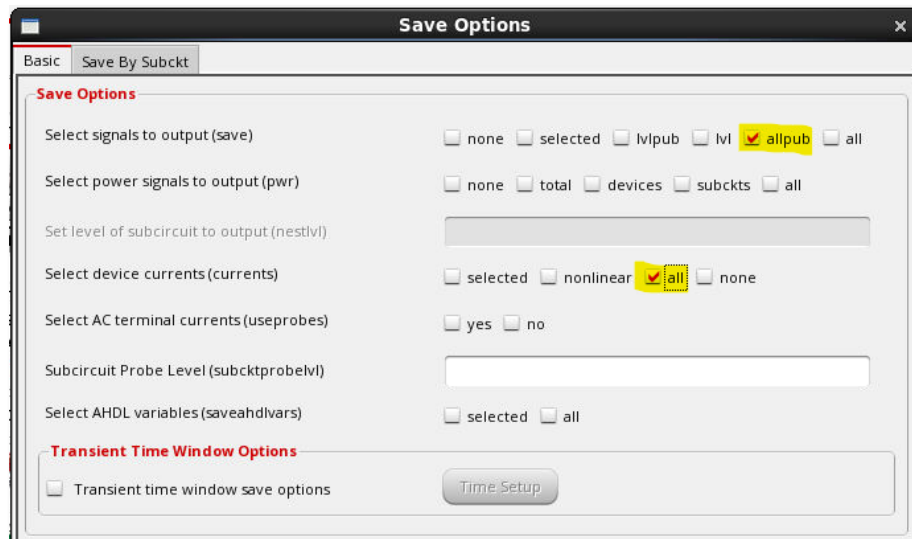


Figure 10: Setting save options

Action 14: In the **ADE Explorer** window, click on **Variables** and select **Copy From Cellview** as shown in Figure 11.

Action 15: Set **avdd** to **2**.

Action 16: Set **avss** to **0**.

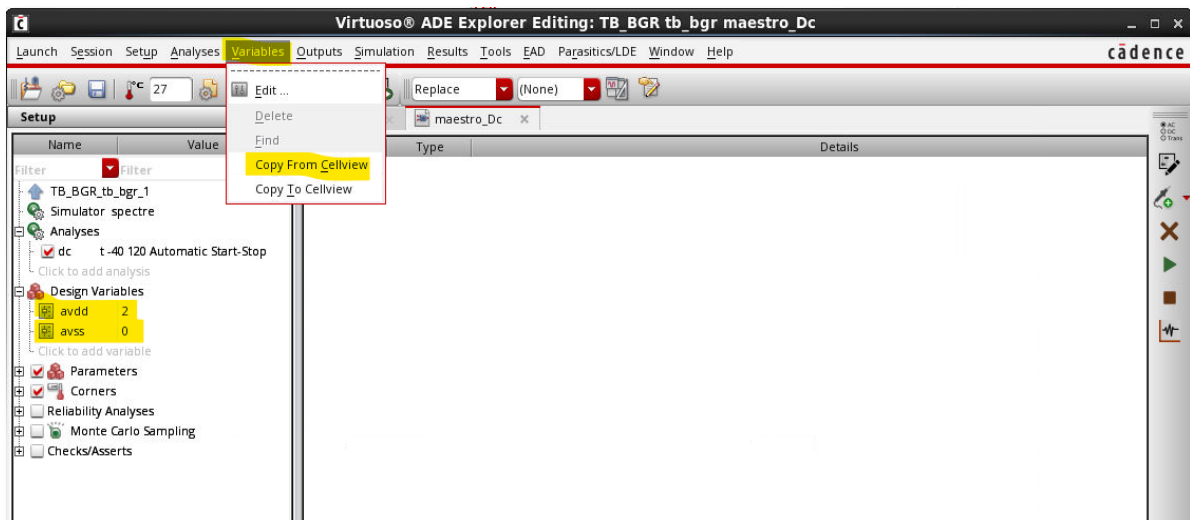


Figure 11: Copying variables from cell view

Action 17: In the **ADE Explorer** window, click on the green arrow button  to **netlist and run** the simulation.

Action 18: In the **ADE Explorer** window, choose **Results > Direct Plot > Main Form** to open the **Direct Plot** form as shown in Figure 12.

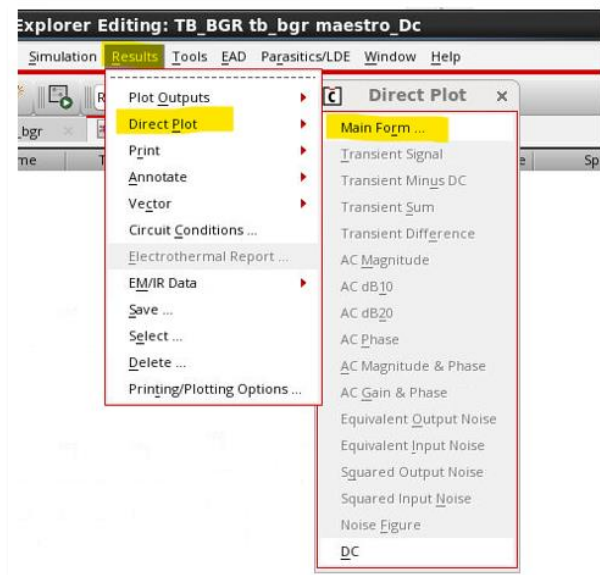


Figure 12: Opening Direct Plot Form from ADE Explorer

Action 19: To plot BGR_output, select **Voltage** as **Function**, **Net** as **Select** from the drop-down list.

Action 20: Enable the **Add To Outputs** checkbox. Select the **V_BGR** net from the schematic window.

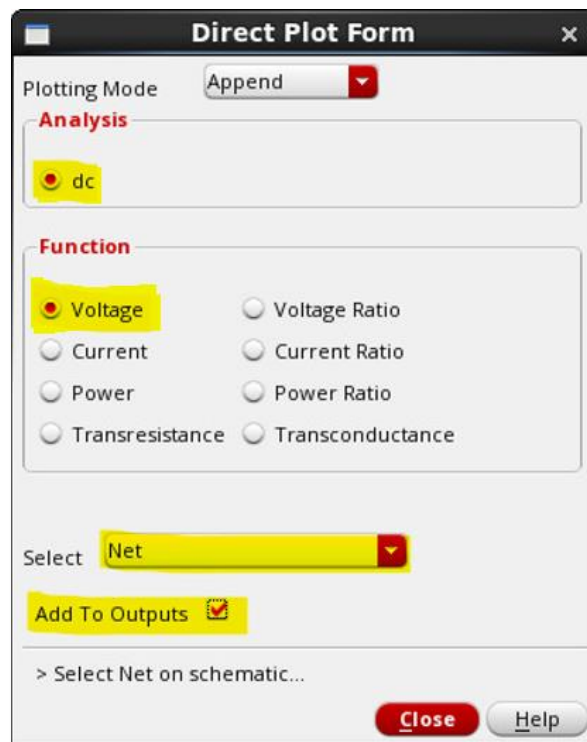


Figure 13: Direct Plot Form

The plot appears as shown in Figure 14. If the waveform is plotted in the docked VIVA assistant within Explorer/Assembler, you can undock it by clicking on .

Action 21: Add a delta marker by hovering over the peak of the plot and pressing the **A** bindkey.

Action 22: Hover over the lowest part of the curve and press the **B** bindkey.

A delta marker will be added like the one shown below.



The `v("/V_BGR" ?result "dc")` expression is also added to the **Outputs Setup** table in the **maestro_Dc** view.

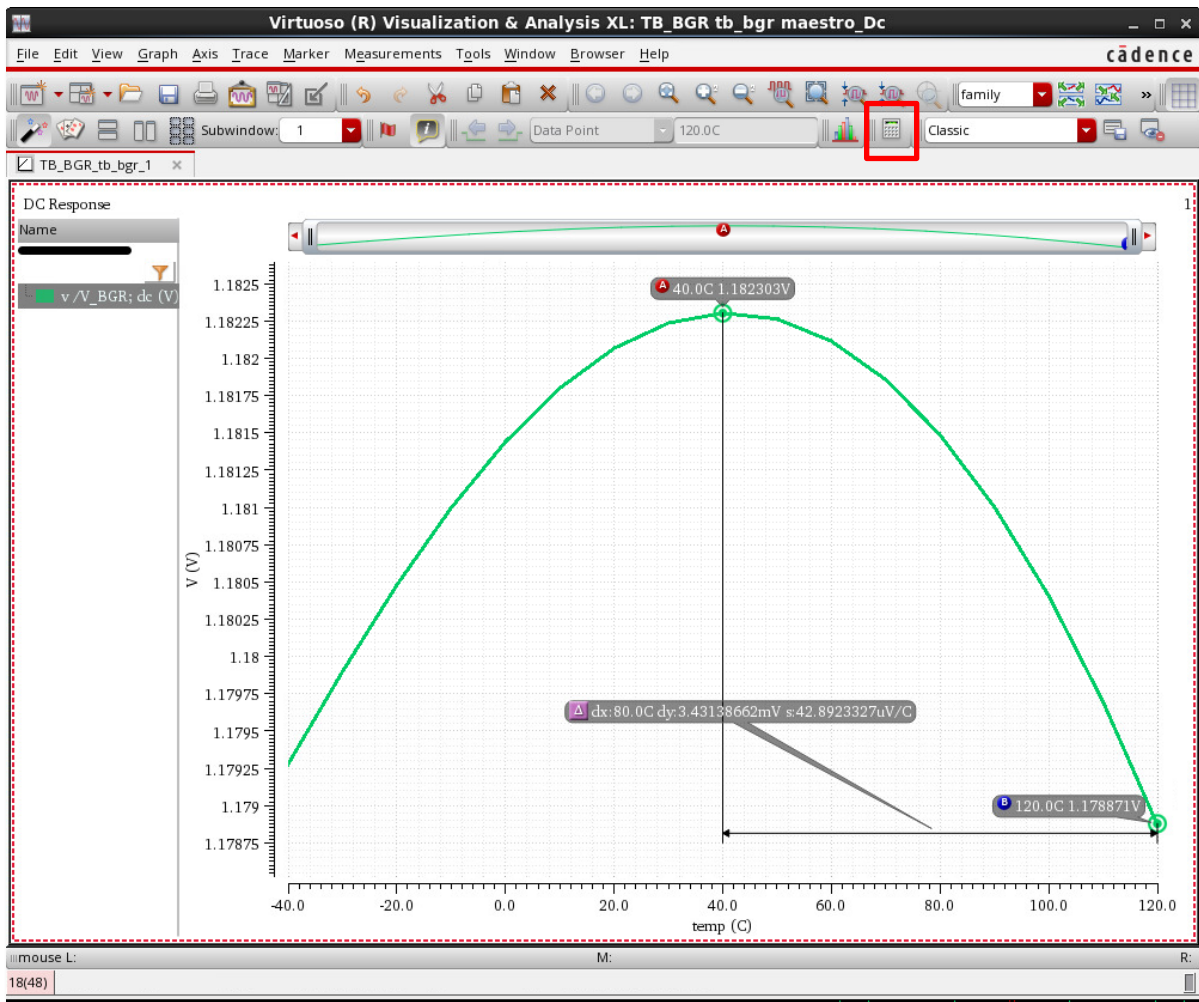


Figure 14: BGR output plot with temperature sweep DC analysis

Action 23: Select the $/V_BGR$ plot on ViVA and click the calculator icon  as shown in Figure 14.

The **ViVA calculator** window will open with the `v("/V_BGR" ?result "dc")` expression as shown in Figure 15.



Figure 15: ViVA-XL calculator window

Action 24: Select the **ymax** function from the **Functional Panel** window.

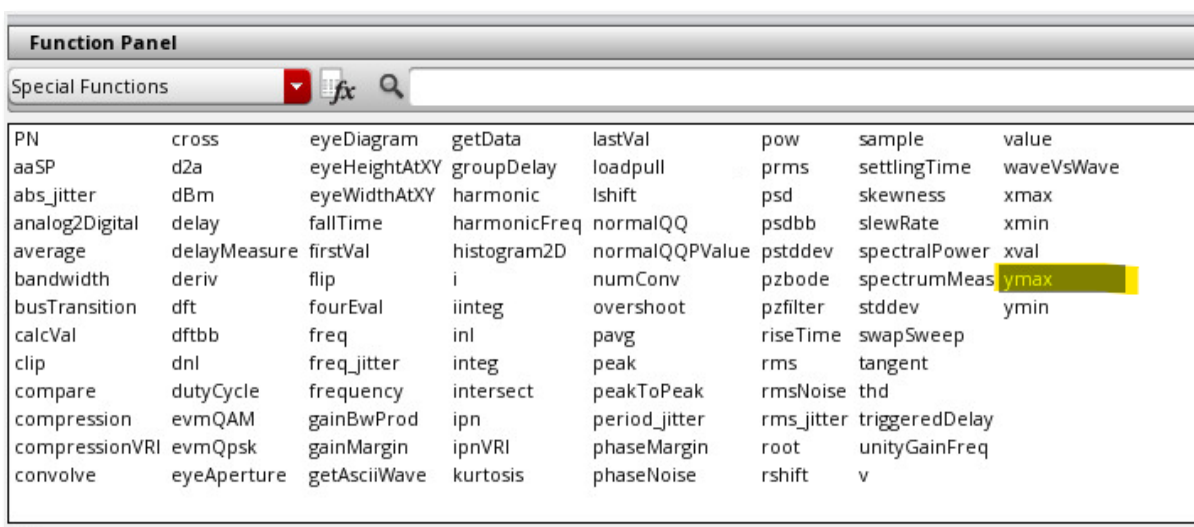


Figure 16: Functional Panel of ViVA-XL calculator

The expression will become `ymax(v("/V_BGR" ?result "dc"))` as shown in Figure 17.

Action 25: In the **ViVA calculator** window, click on the **Send buffer expression to ADE** output icon .

The expression is added to the **ADE Output Setup** table.



Figure 17: ViVA-XL calculator buffer

Action 26: Repeat this to add the following expression to the **ADE Output Setup** table:

```
ymin(v("/V_BGR" ?result "dc"))
```



Figure 18: ViVA-XL calculator buffer

Action 27: In the **ViVA calculator** window, write the following expression:

```
ymax(v("/V_BGR" ?result "dc")) - ymin(v("/V_BGR" ?result "dc"))
```

Action 28: Add this expression in the **ADE maestro** view using the **Send buffer expression to ADE** output  icon.

Action 29: Evaluate the expression value using the **Evaluate the buffer**  icon.

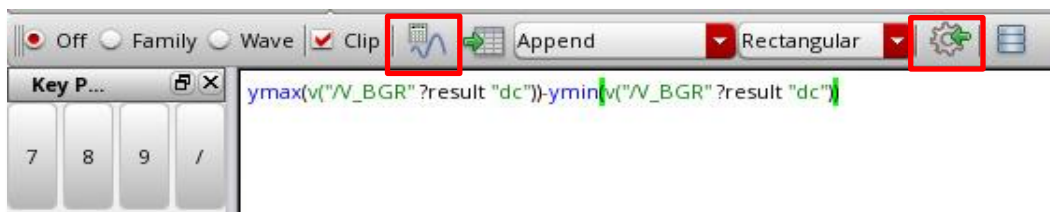


Figure 19: ViVA-XL calculator buffer

Action 30: In the **ADE Explorer** window, double-click on the **Name** column for the `v("/V_BGR" ?result "dc")` expression and type in the name `BGR_output` as shown in Figure 20.

Action 31: Similarly, add the name `BGR_variation` for the last expression.

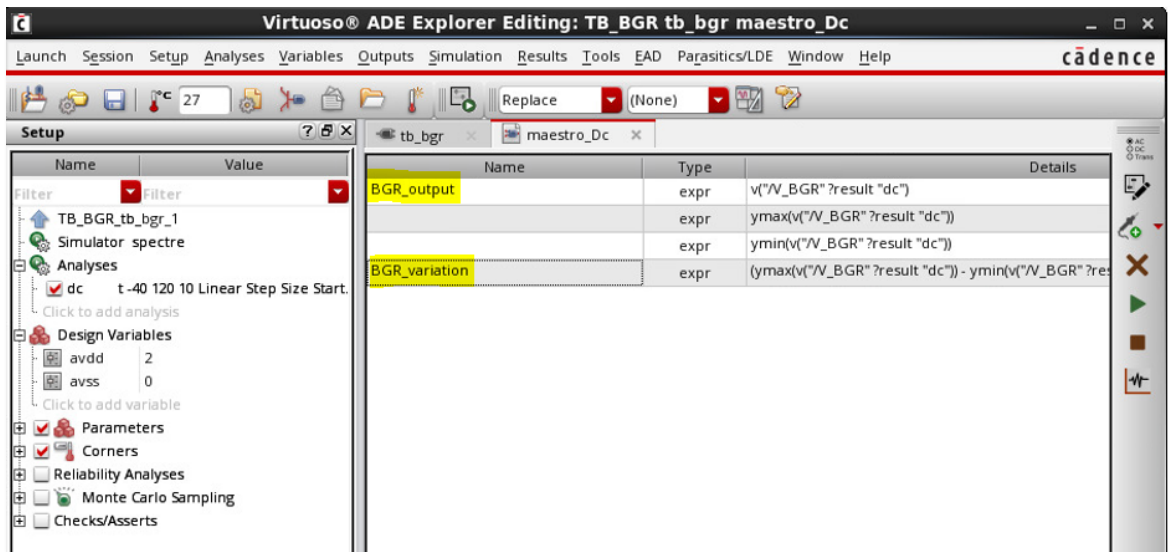


Figure 20: ADE maestro_Dc view

Action 32: Close the **ViVA** window.

Action 33: In the **ADE Explorer** window, click on the **Add outputs** icon  and select **OP Parameters**.

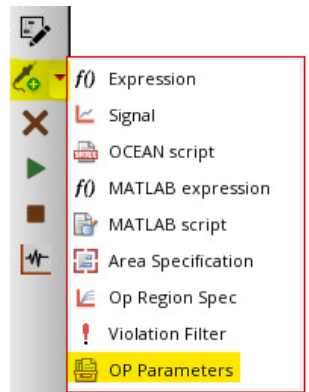
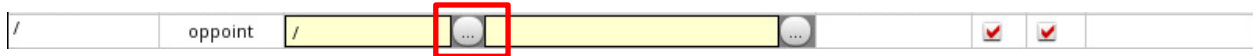


Figure 21: Adding operating point parameters in output

This adds an operating parameter expression row.

Action 34: Click on the **select operating point instances icon**  to open the schematic.

Action 35: Descend into the `BGR_circuit` schematic and select the **M12** pmos instance.



This adds the **/I18/M12** instance.

Action 36: Now, click on the **Select operating point parameters** icon to open the **Select OP Parameters** form.

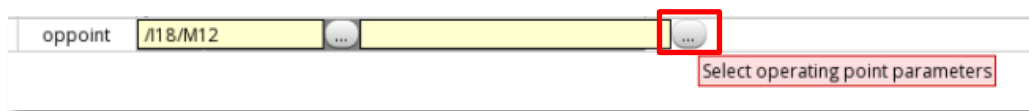


Figure 22: Operating point in output

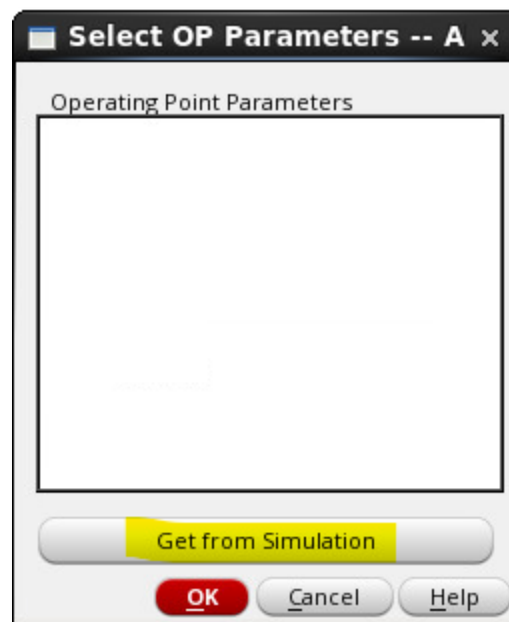


Figure 23: Select operating point parameter for /I18/M12 instance

Action 37: Click on **Get from Simulation**.

All operating points associated with the **/I18/M12** instance will appear in the **Select OP Parameter** form.

Action 41: In the **Direct Plot Form** window, select **Current** as **Function** and enable the **Add To Outputs** checkbox.

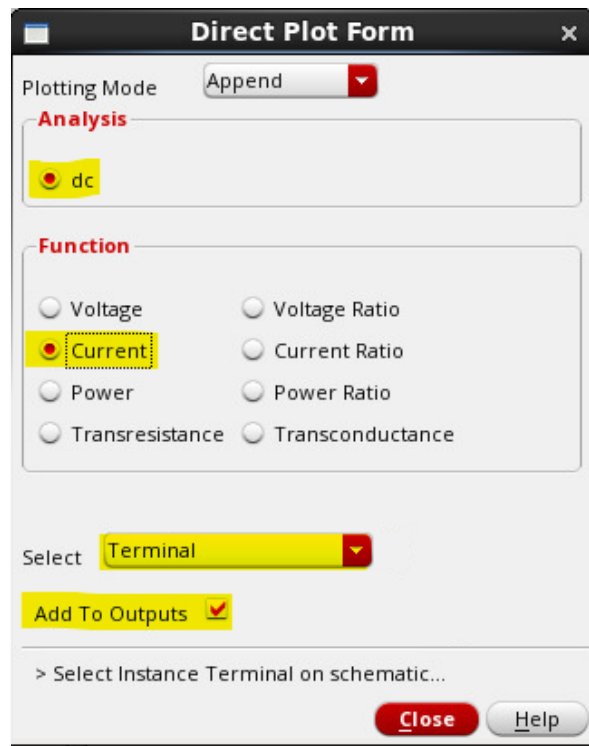


Figure 27: Direct Plot Form

Action 42: Go to the schematic tab and return to the level above by clicking the up arrow on the schematic Go toolbar.

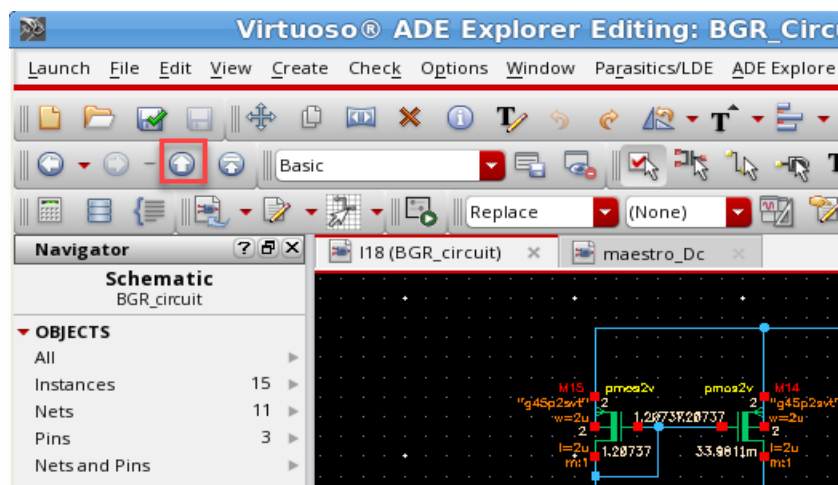


Figure 28: Clicking the up arrow on the schematic

Action 43: Select the **PLUS** node of the **V7** supply voltage source as shown in Figure 29.

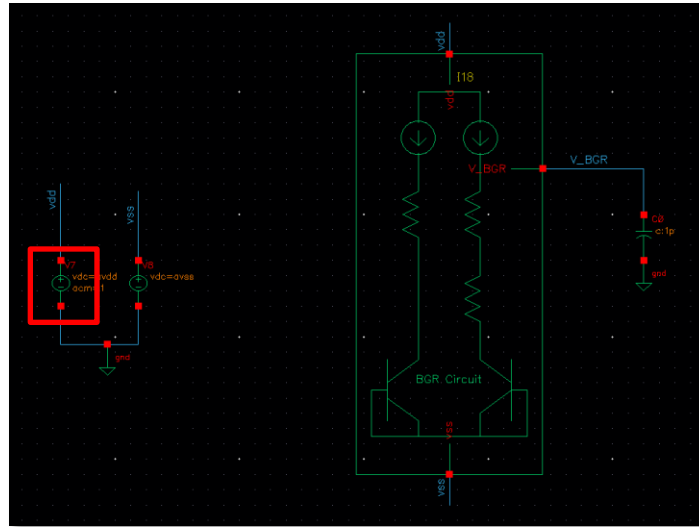


Figure 29: Plotting supply current

The waveform is plotted as shown in Figure 30. The `"i("/V7/PLUS" ?result "dc")` supply current expression is added to the **Outputs Setup** table in ADE Explorer.

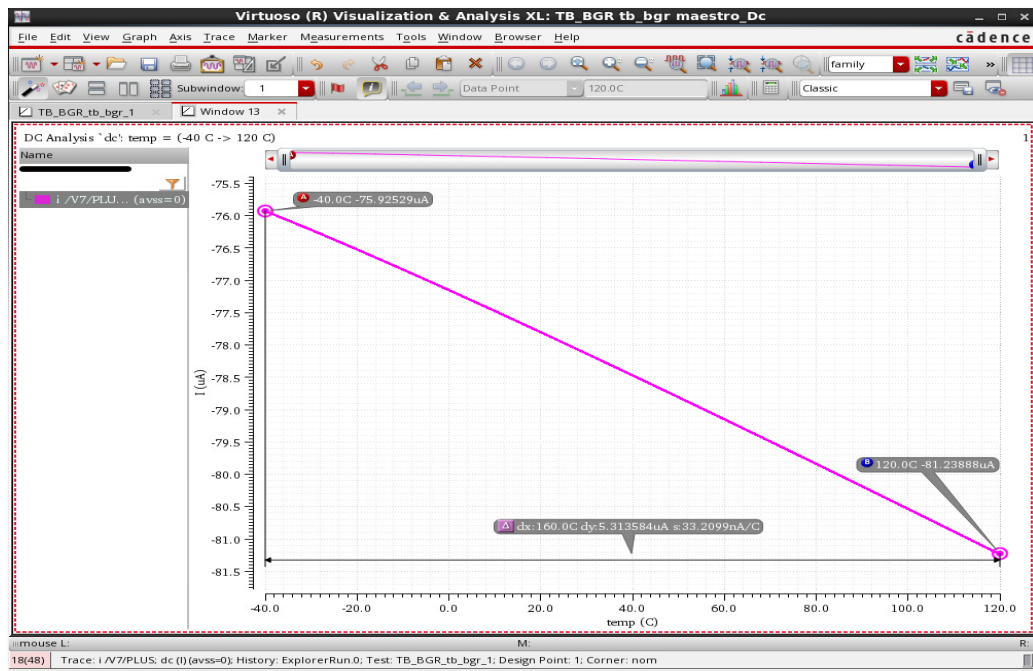


Figure 30: BGR supply current plot with temperature sweep DC analysis

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Action 44: Name this expression as `Supply_Current` in the ADE Outputs Setup table.

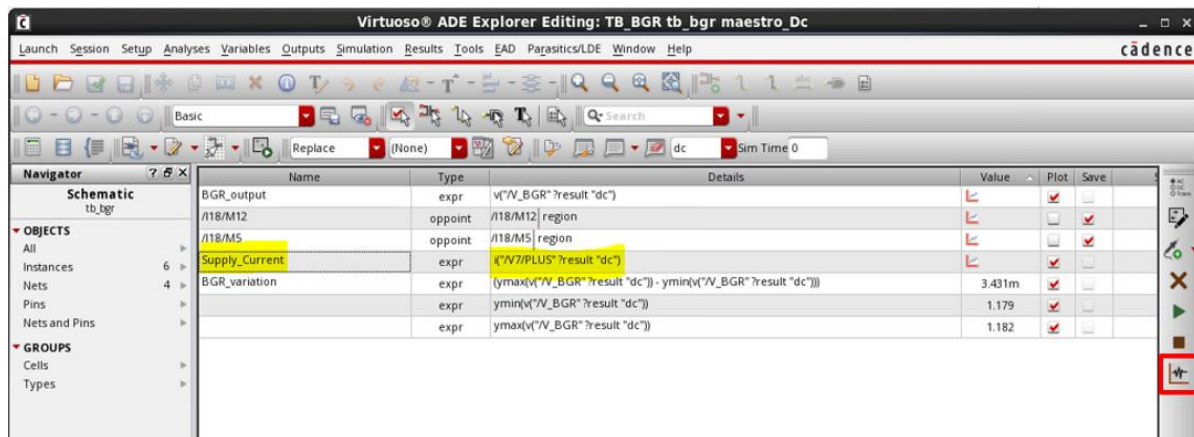


Figure 31: ADE maestro_Dc view

Action 45: In the ADE Explorer window, click on the **Plot Outputs** icon  to see all plots together as shown in Figure 32.

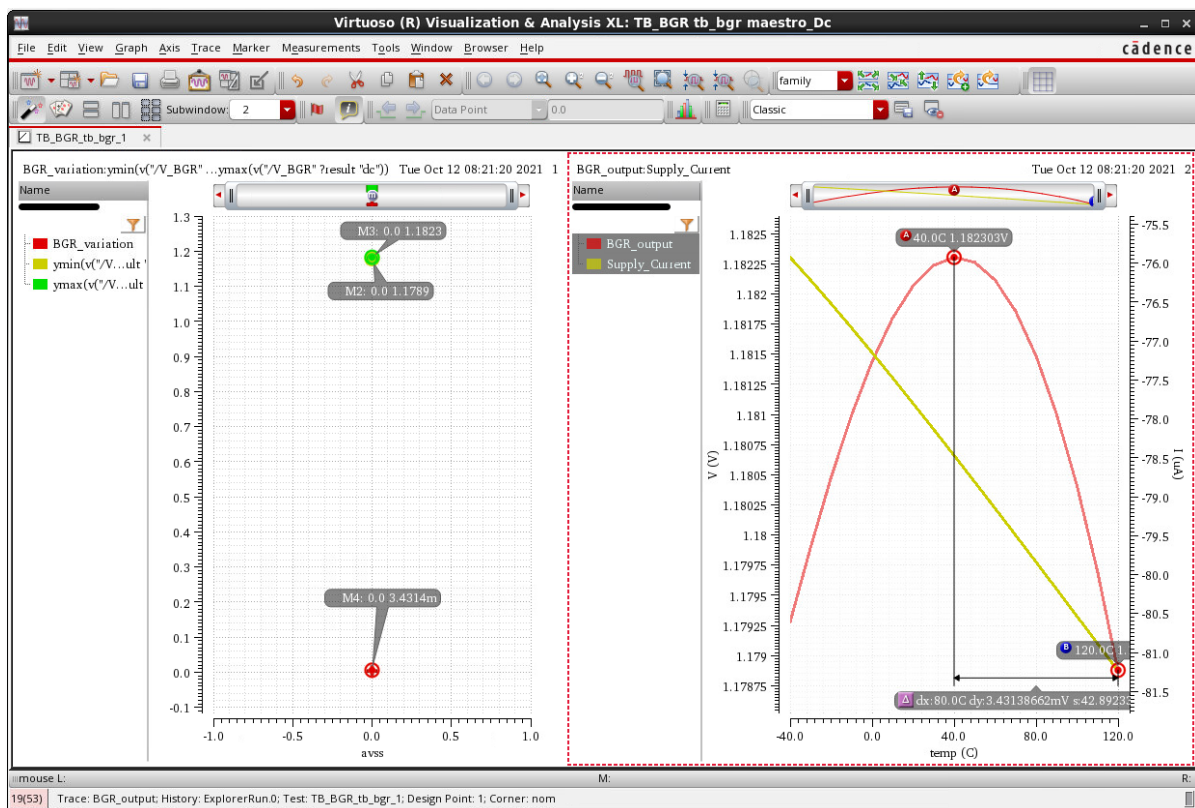


Figure 32: Plots in ViVA

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Action 46: In the **ADE Explorer** window, click on the up arrow to open the **ADE Assembler** window as shown in Figure 33.

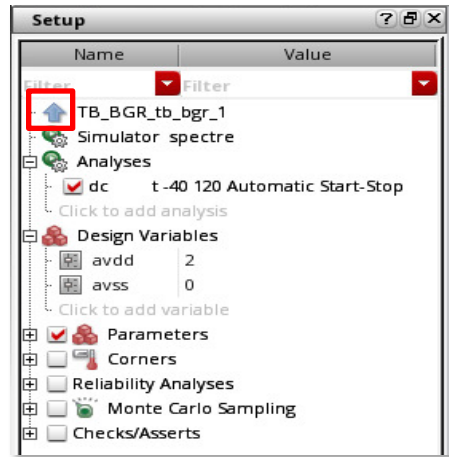


Figure 33: Up Arrow to open ADE Assembler

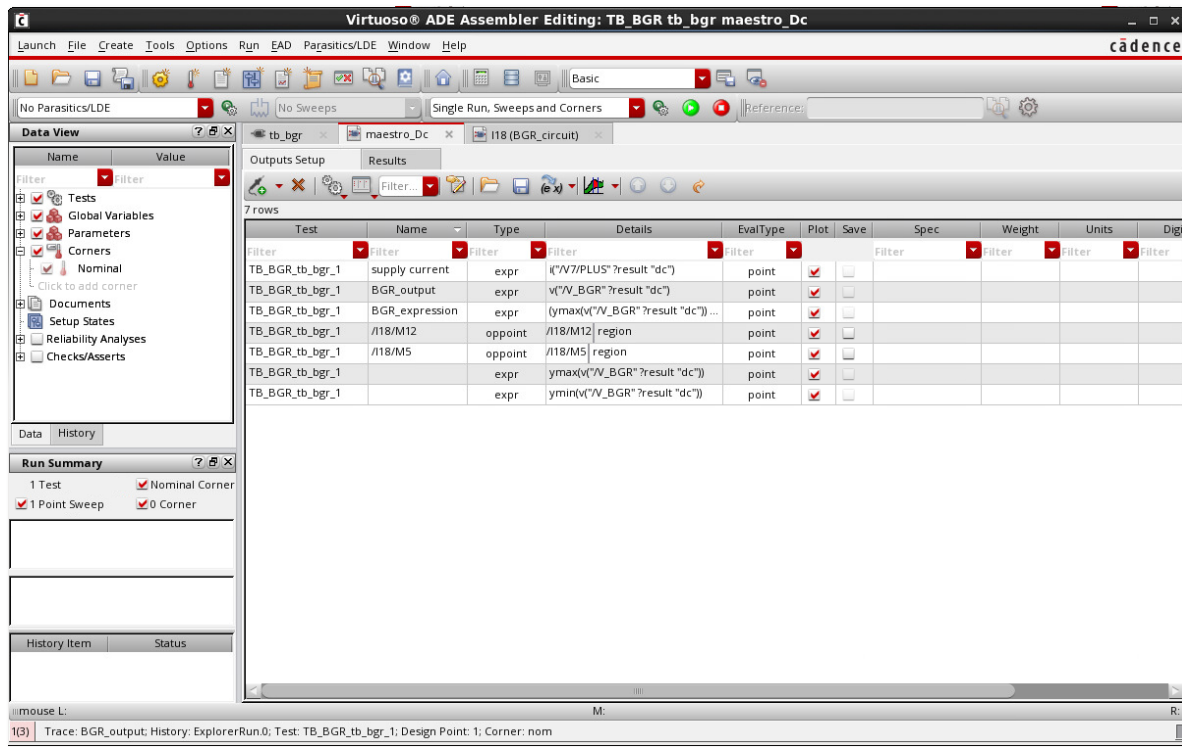


Figure 34: Assembler view of maestro_Dc

Action 47: In the ADE Assembler **Data View** window, expand **Corners** using the + sign and then, **Click to add corners** as shown in Figure 35.

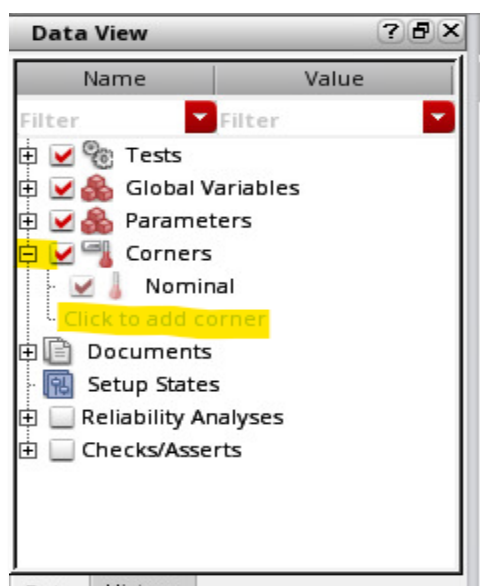


Figure 35: Opening Corner Setup

The **Corners Setup** window will open.

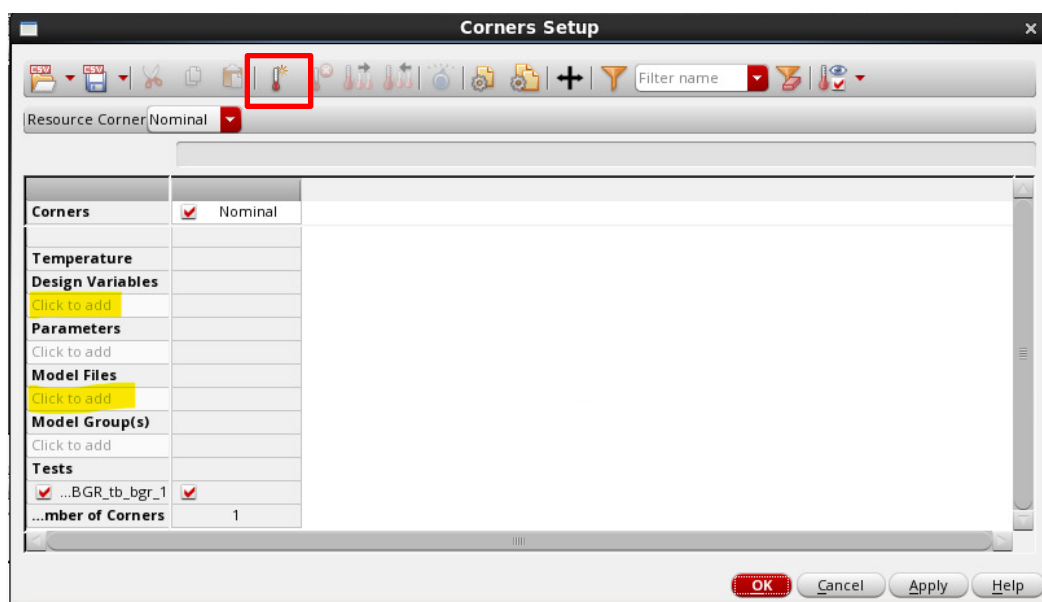


Figure 36: Initial Corner Setup

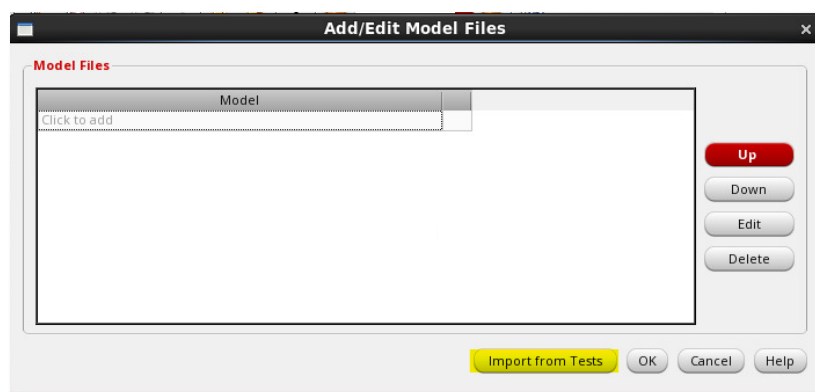
Action 48: Click on the **add new corner** icon . This will add a corner, **C0**.

Action 49: Click on **Click to add** under **Design Variables** as shown in Figure 37. Select **avdd** and then enter the values 1.8 2.2.

Corners	☒ Nominal	☒ CO
Temperature		
Design Variables		
Click to add		
Parameters		
Click to add		
Model Files		
Click to add		
Model Group(s)		<modelgroup>
Click to add		
Tests		
☒ ...BGR_tb_bgr_1	☒	☒
Tags		

Figure 37: Adding new corner, design variables, and model files in corner setup

Action 50: Click on **Click to add** under **Model Files** as shown in Figure 38.



The dialog box titled "Add/Edit Model Files" contains a table with a header "Model" and a row with "Click to add". To the right of the table are buttons: "Up" (red), "Down", "Edit", and "Delete". At the bottom are buttons: "Import from Tests" (yellow), "OK", "Cancel", and "Help".

Figure 38: Add/Edit Model Files form

Action 51: Click on **Import from Tests** and then, click **OK**.

Action 52: Select the **tt**, **ss**, **sf**, **fs**, **ff** sections and ensure the box is checked. Then, click **OK**.

The **Corners Setup** form will look like Figure 39.

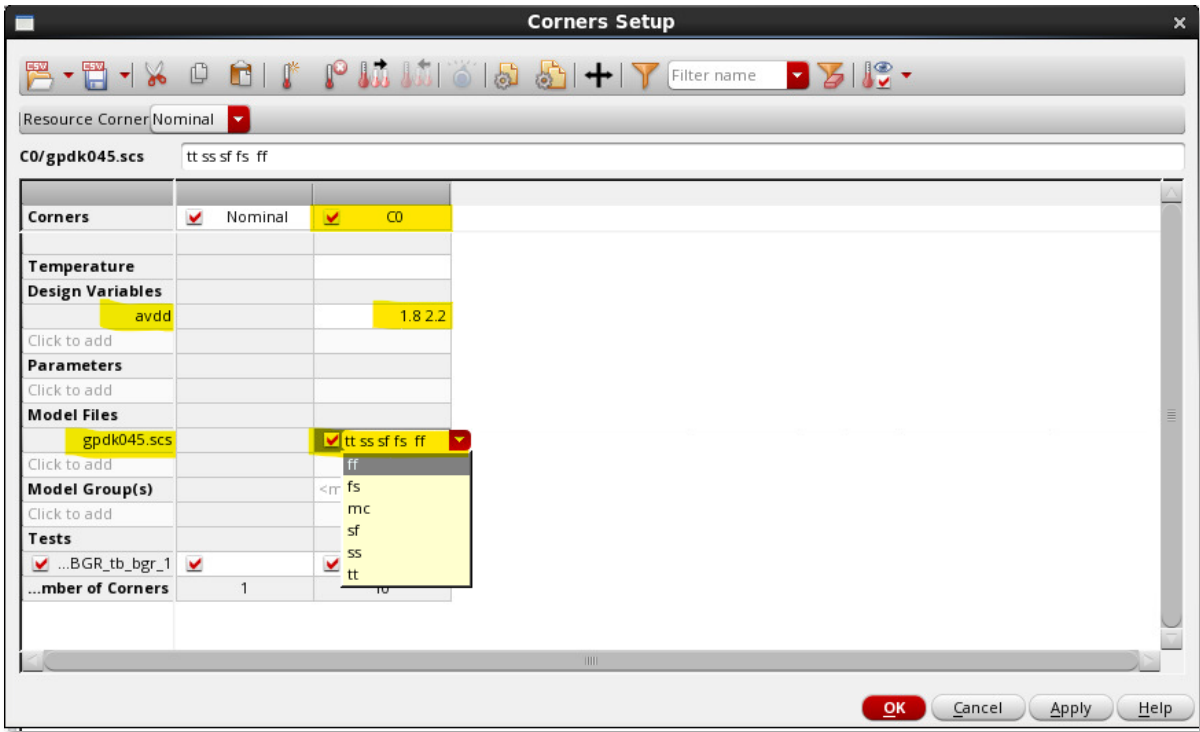


Figure 39: Final Corner Setup

Action 53: To add specifications to the **Outputs Setup** of the **ADE Assembler** view, double-click on the **Spec** field on the row for the **y_{max}** expression.

Select **<** from the drop-down menu. Add the **1.195** value as shown in Figure 40.

Test	Name	Type	Details	EvalType	Plot	Save	Spec	Weight
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
TB_BGR_tb_bgr_1	BGR_output	expr	v("/V_BGR"?result "dc")	point	☒	☐		
TB_BGR_tb_bgr_1	BGR_variation	expr	(ymax(v("/V_BGR"?result "dc")) - ymin(v("/V...	point	☒	☐		
TB_BGR_tb_bgr_1		expr	ymin(v("/V_BGR"?result "dc"))	point	☒	☐		
TB_BGR_tb_bgr_1		expr	ymax(v("/V_BGR"?result "dc"))	point	☒	☐	<	
TB_BGR_tb_bgr_1	/I18/M12	oppoint	/I18/M12 region	point	☐	☒	none	
TB_BGR_tb_bgr_1	/I18/M5	oppoint	/I18/M5 region	point	☐	☒	minimize	
TB_BGR_tb_bgr_1	Supply_Current	expr	i("/V7PLUS"?result "dc")	point	☒	☐	>	
							range	
							tol	
							info	
							wave	

Figure 40: Adding specs in Outputs Setup of ADE Assembler

Action 54: Repeat this using the following table to add specifications to the other expressions.

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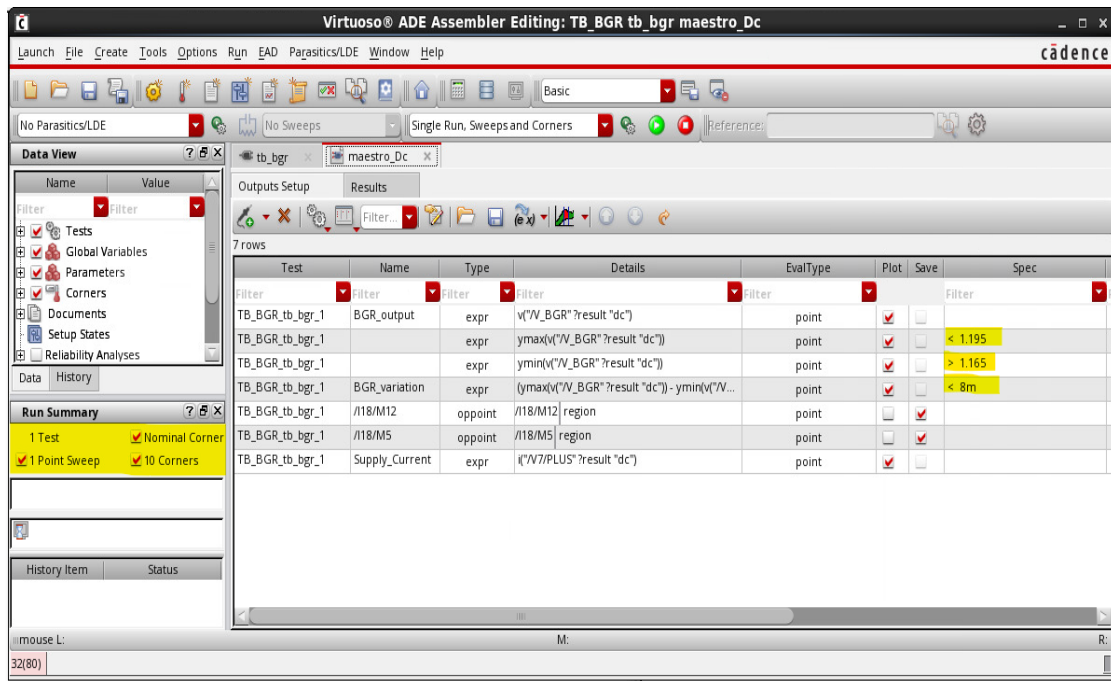


Figure 41: ADE Assembler setup of maestro_Dc

Action 55: Run the simulation by clicking on the green run icon  to **netlist and run simulation** across corners.

All output design parameters, BGR_variation, supply current, ymax, and ymin of BGR_output are passing the specifications.

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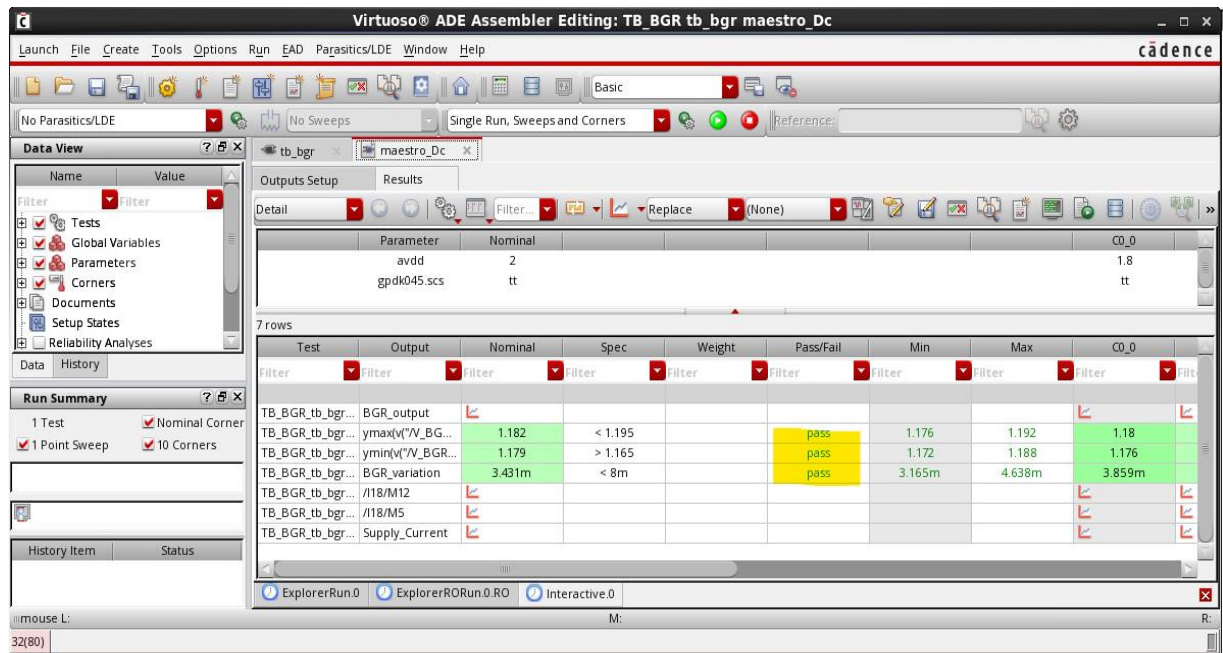


Figure 42: Results tab view of Assembler maestro_Dc

Action 56: Right-click on the **Nominal** column for BGR_output and select **Plot Across Corners** as shown in Figure 43.

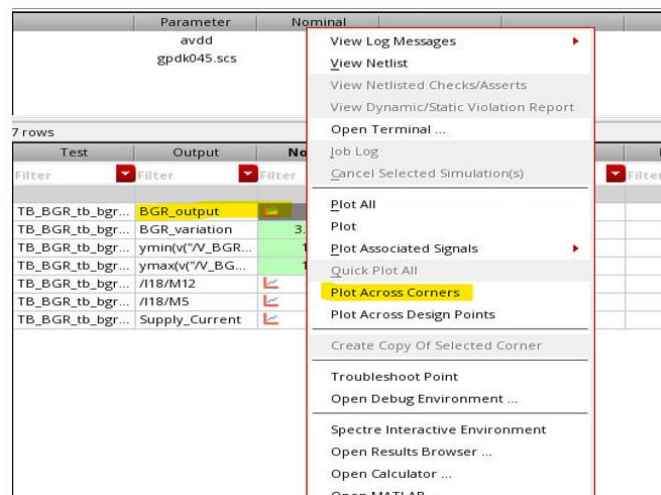


Figure 43: Plotting across corners

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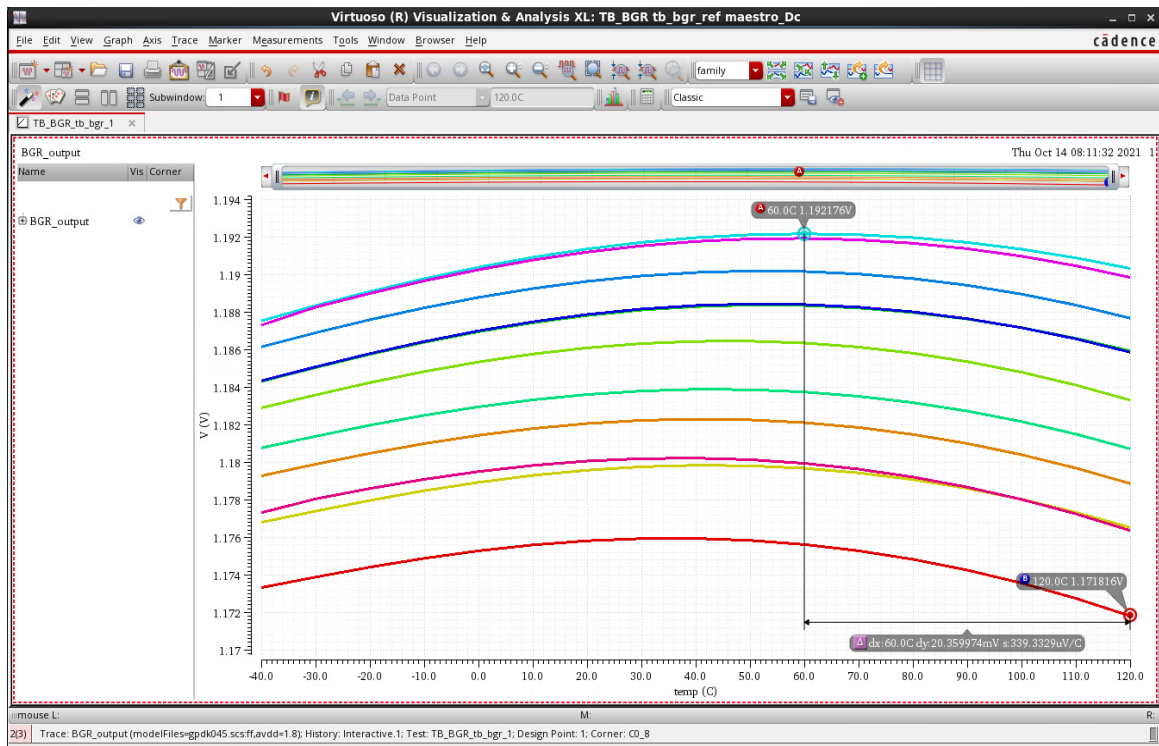


Figure 44: BGR_output plots Across PVT Corners

Action 57: Close all windows but leave the **Virtuoso** session open for the next lab.

Lab 2: Power Supply Rejection Ratio of Bandgap Using Ac Analysis

This lab is set up to determine PSRR of the Bandgap circuit. PSRR is a measure of the influence of the power supply ripple on the Bandgap output voltage.

PSRR analysis is used to measure the effect of supply ripples on the output reference supply.

Action 58: From the Library Manager, open **TB_BGR > tb_bgr > schematic**. The schematic window opens as shown in Figure 45.

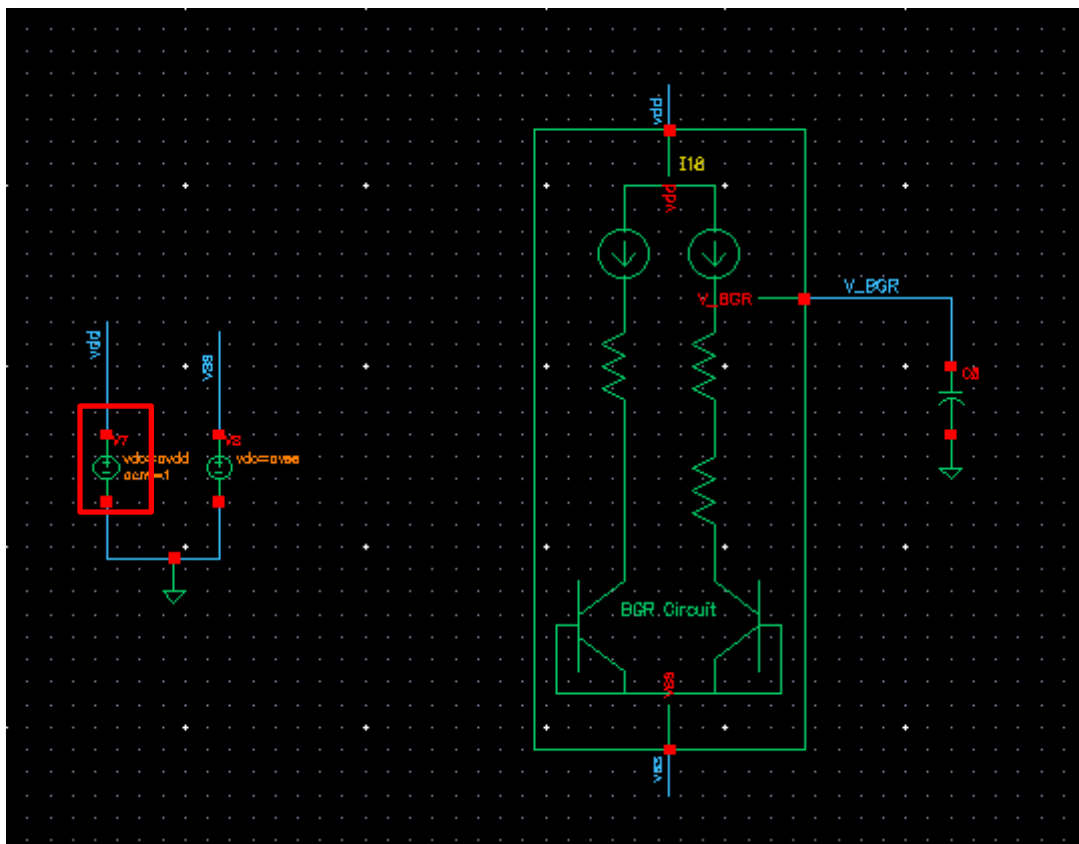


Figure 45: Schematic of tb_bgr cell testbench

Action 59: In the **schematic** view, select the **V7** dc supply voltage source and press the **Q** bindkey key to open the **Edit Object Properties** form. Confirm that **AC magnitude** is defined as **1V**.

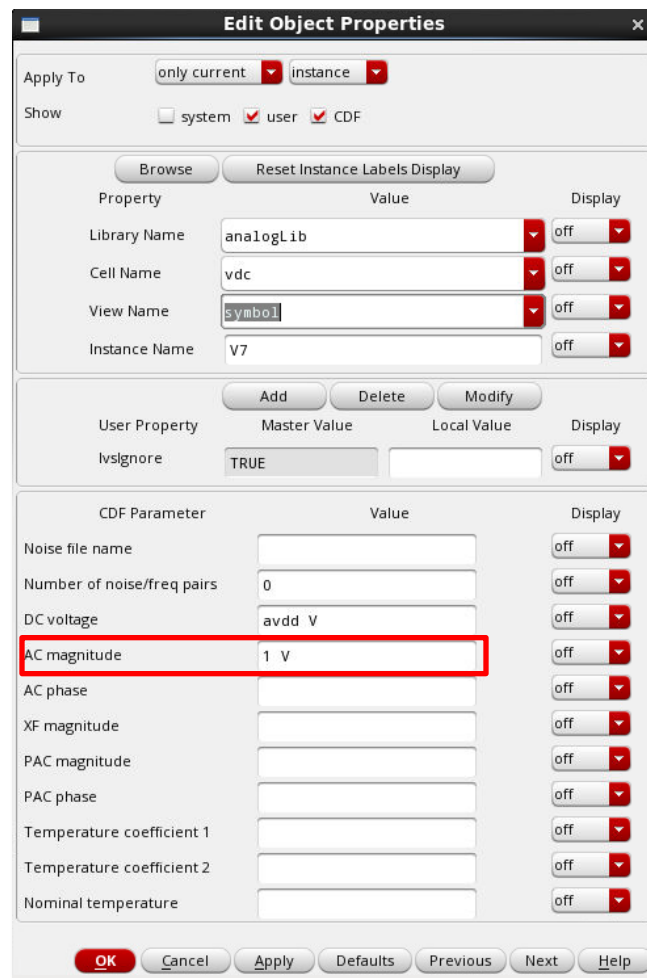


Figure 46: Object property of supply voltage source

Action 60: Cancel the form.

Action 61: From the schematic, select **Launch > ADE Explorer**.

Action 62: In the **Launch ADE Explorer** window that opens, select **Create New View** and click **OK**.

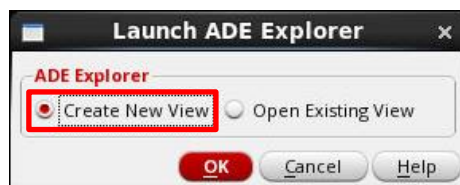


Figure 47: Launching ADE Explorer from the schematic window

Action 63: In the **Create new ADE Explorer view** window, change the view name to `maestro_Ac`.

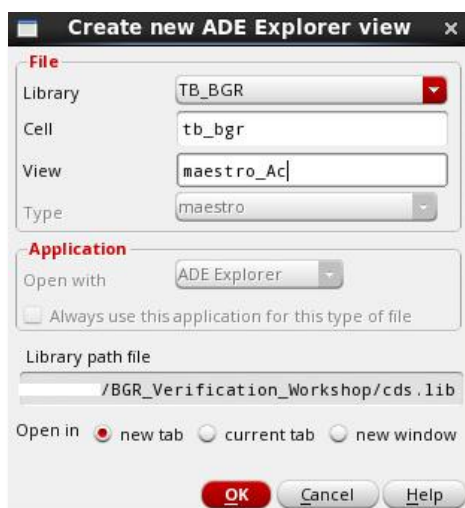


Figure 48: Opening maestro_Ac view of tb_bgr cell

The **ADE Explorer** window opens as shown in Figure 49.

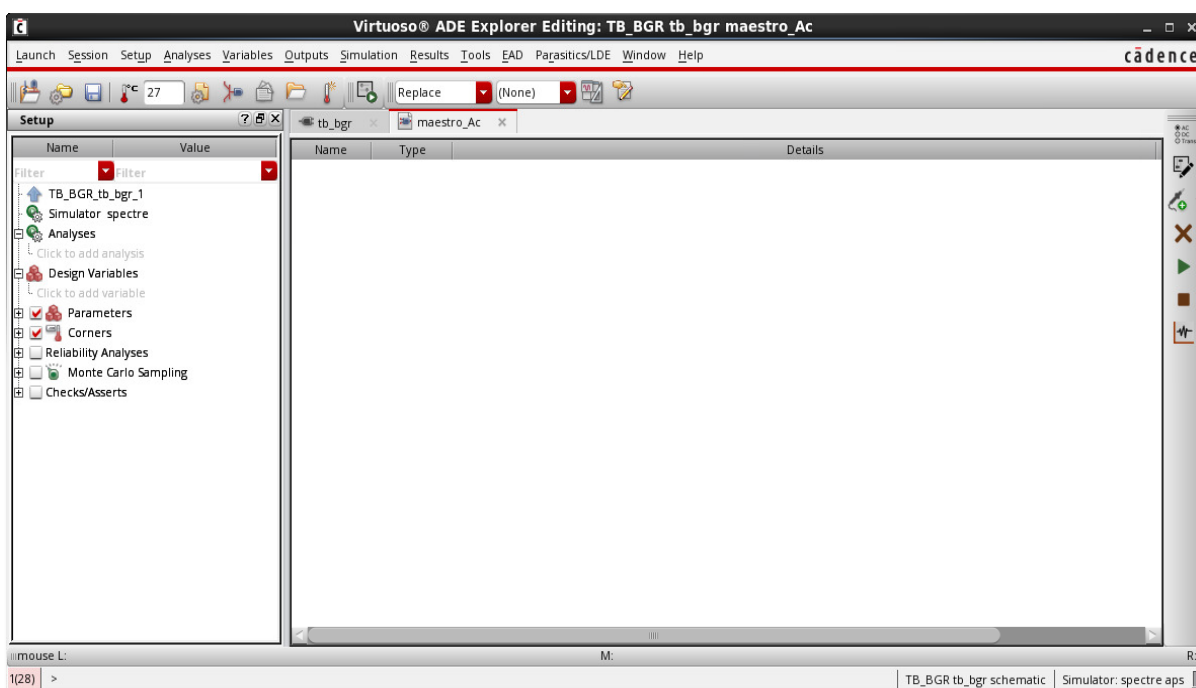


Figure 49: ADE Explorer window showing maestro_Ac view

Action 64: In the **ADE Explorer** window, select **Click to add analysis** in the **Analyses** section of the **Setup** assistant.

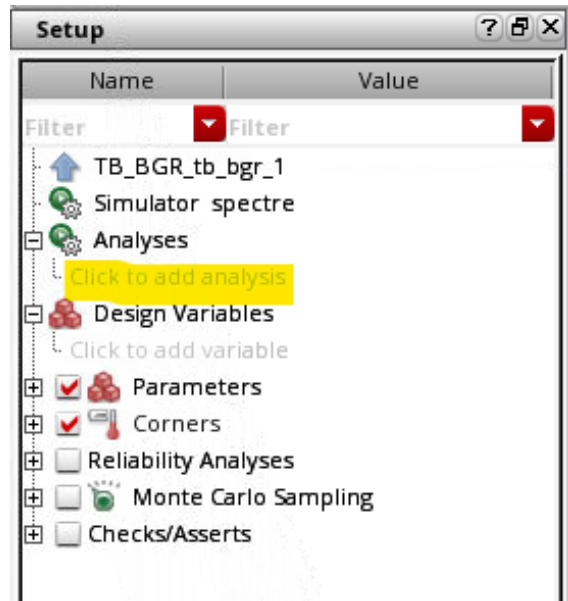


Figure 50: Opening 'Analyses' form from 'Click to add analysis'

Action 65: In the **Choosing Analyses** form, select the **ac** analysis and set the following options. Then, click **OK**:

- Select **Frequency** as **Sweep Variable**.
- Set the **Start** frequency to **1Hz** and **Stop** frequency for **1GHz**.
- Choose **Automatic** for **Sweep Type**.

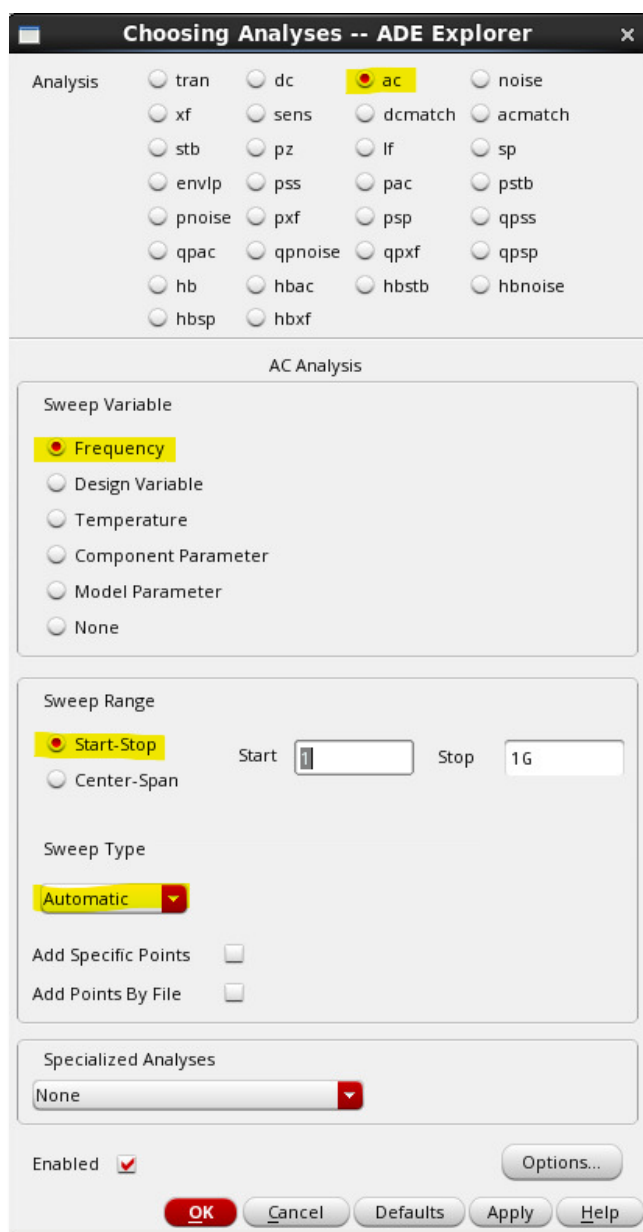


Figure 51: Setup of DC-Sweep Analysis

Action 66: In the **ADE Explorer** window, click on **Variables** and select **Copy From Cellview** as shown in Figure 52.

Action 67: In the **Setup** assistant, set the **Design Variables** values for **avdd=2** and **avss=0**.

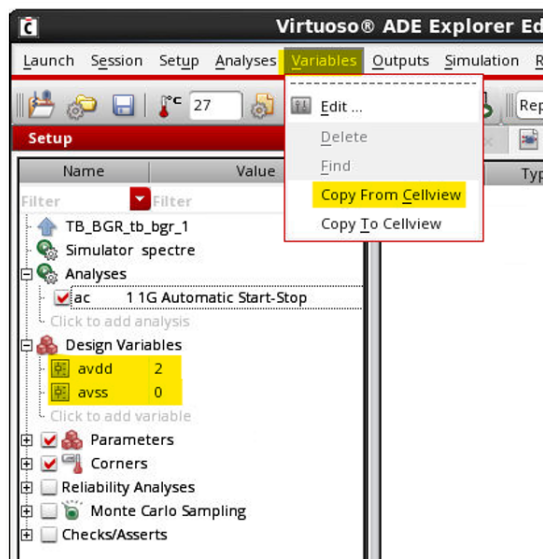


Figure 52: Copying variables from cell view

Action 68: In the **ADE Explorer** window, click on the green arrow button  to **netlist and run** the simulation.

Action 69: When the simulation is complete, click on **Simulation > Output Log** in the **ADE Explorer** window as shown in Figure 53.

Action 70: Verify that simulation has completed with 0 errors.

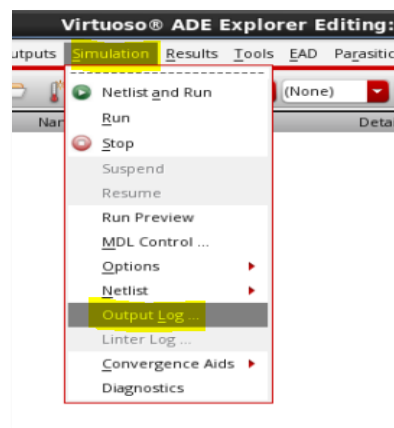


Figure 53: Opening simulation log file from ADE Explorer window

Action 71: In the **ADE Explorer** window, choose **Results > Direct Plot > Main Form** to open **Direct Plot Form** as shown in Figure 54.

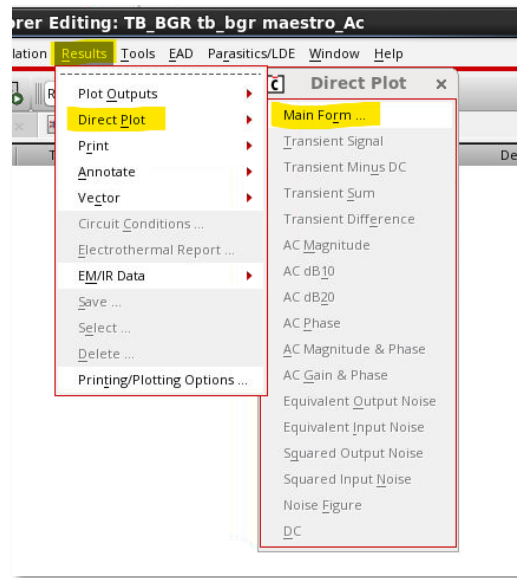


Figure 54: Opening Direct Plot Form from ADE Explorer window

Action 72: To plot the gain, select **Voltage** as the **Function**, **Net** from the drop-down menu, and **dB20** as **Modifier**.

Action 73: Enable the **Add To Outputs** checkbox and select the **V_BGR** net from the schematic.

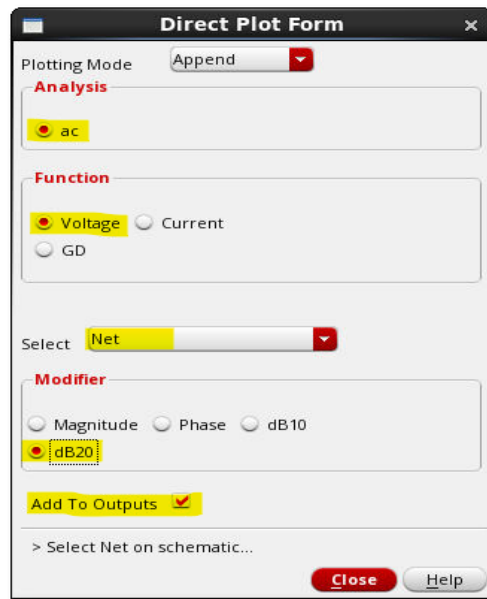


Figure 55: Direct Plot Form

Bandgap Reference Verification: RAK

The PSRR waveform is plotted as shown in Figure 56. The `db(vfreq('ac'"/V_BGR"))` expression is added to the maestro view.

Action 74: Add single-point markers using the **M** bindkey on the plot.

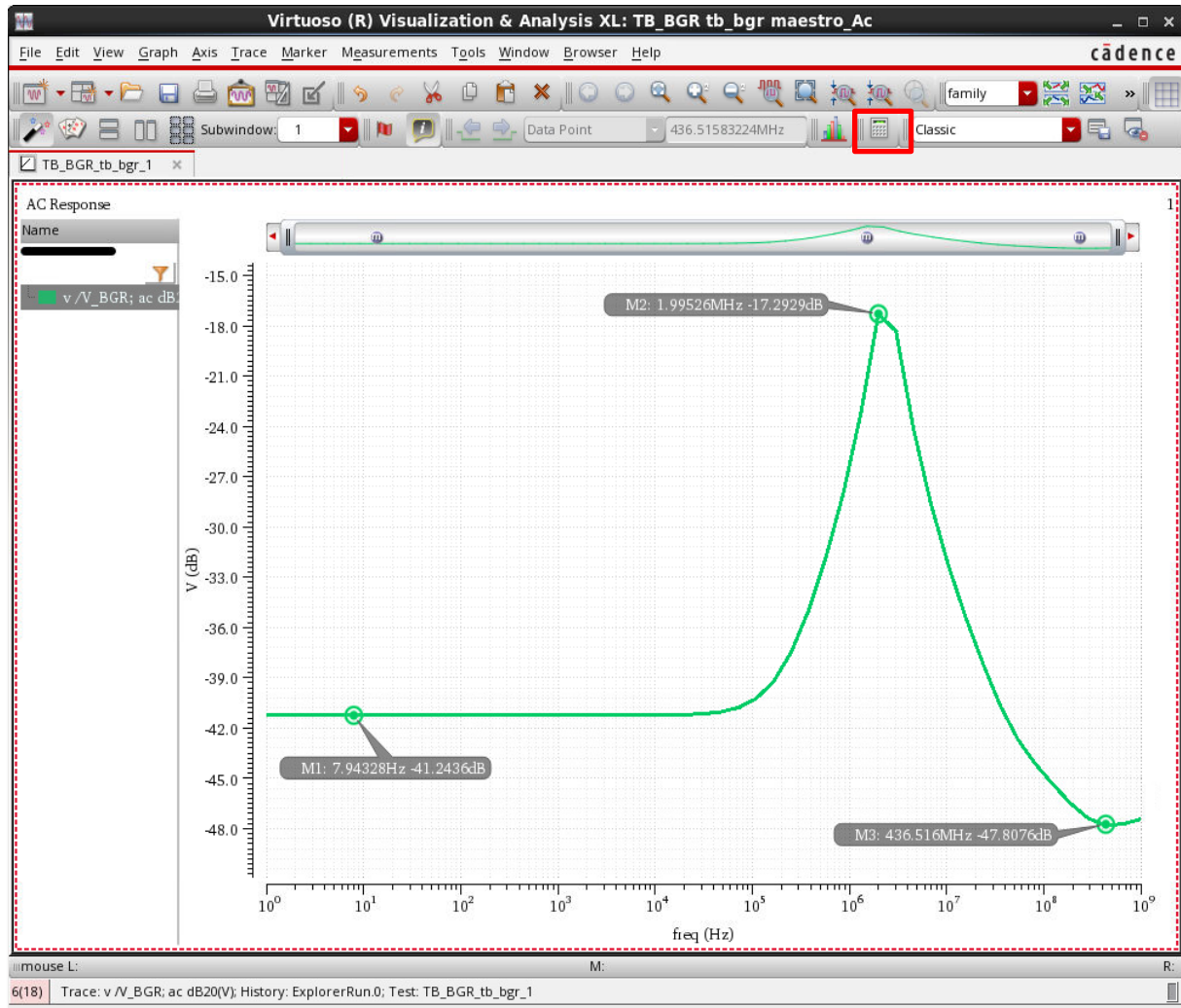


Figure 56: Plot of PSRR in ViVA-XL waveform window

Action 75: Select the PSRR plot and click the **calculator** icon  as shown in Figure 56.

The expression for the PSRR plot is added to the calculator as shown in Figure 57.



Figure 57: ViVA-XL calculator

Action 76: In the calculator, select the **value** function from the **Functional Panel** window.

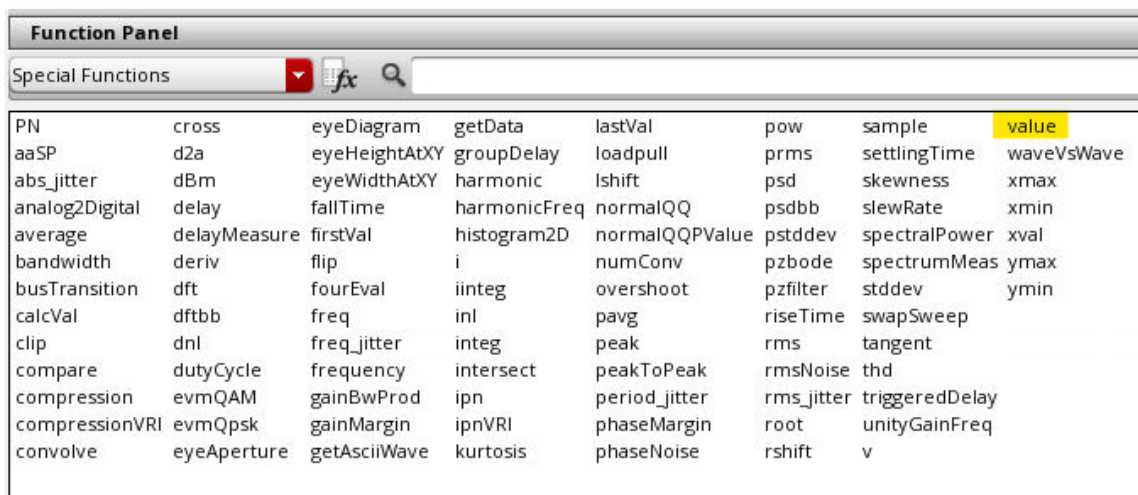


Figure 58: Functional Panel of ViVA-XL calculator

Action 77: The **Functional Panel** window will contain the signal name as shown in Figure 59. Set the value of **Interpolate at** as **1**. Click **OK**.

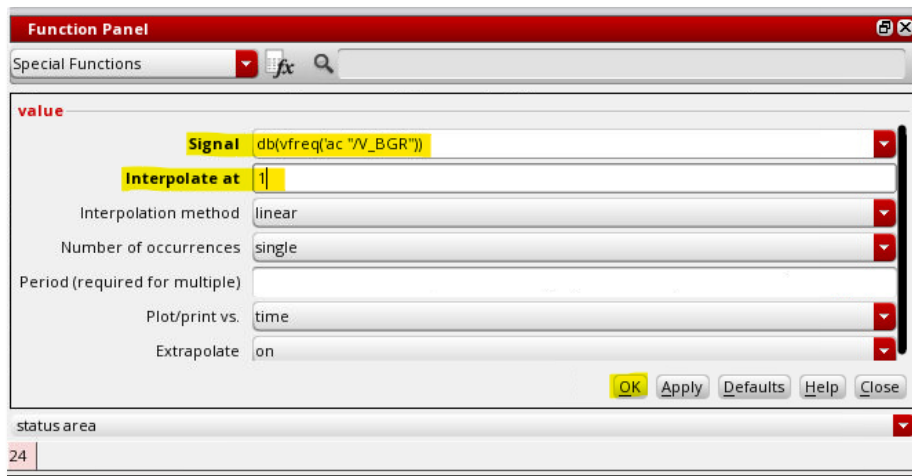


Figure 59: Using value calculator function of ViVA-XL calculator

The expression will become `value(db(vfreq('ac "/V_BGR")) 1 )` as shown in Figure 60.

Action 78: Click the **Send buffer expression to ADE output** icon  to add the expression to the maestro view.

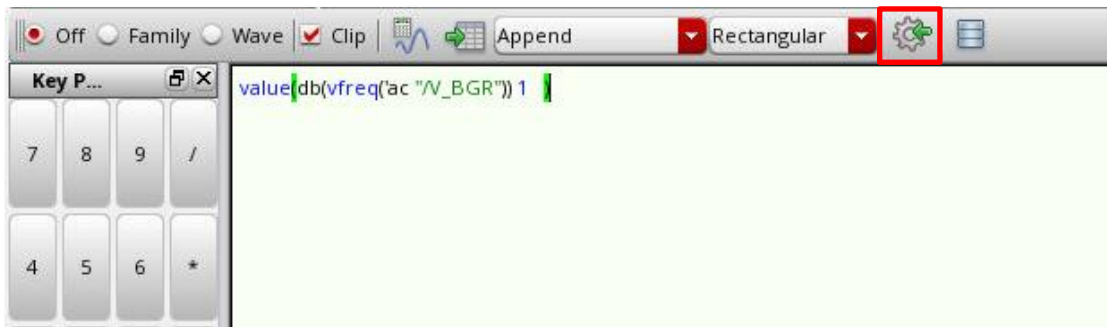


Figure 60: ViVA-XL calculator buffer

Action 79: In the **ADE Explorer** window, select the **Name** field and add the expression name `BGR_output` as shown in Figure 61.

Action 80: Name the first expression as `PSRR_plot`.

Bandgap Reference Verification: RAK

The maestro view should now look like this:

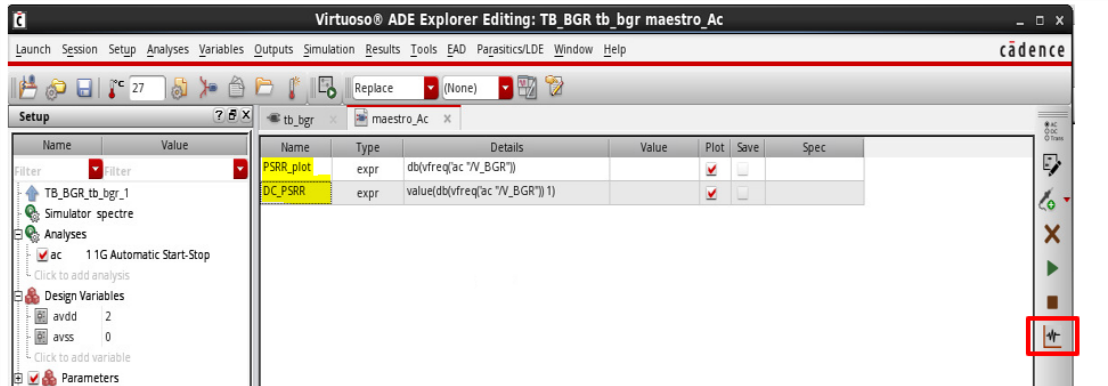


Figure 61: ADE maestro_Ac view

Action 81: Click **Plot Outputs** . The expressions are plotted as shown in Figure 62.

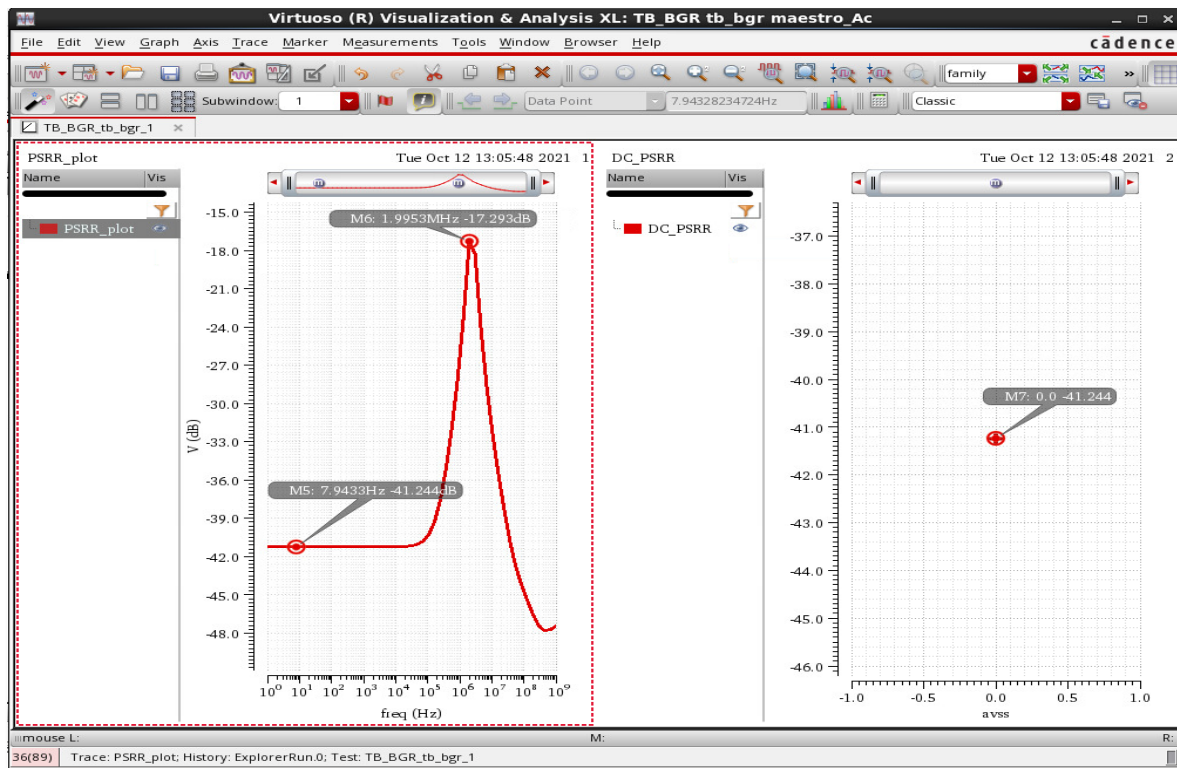


Figure 62: Output plots

Action 82: Close all windows but leave the **Virtuoso** session open for the next lab.

Lab 3: Loop Gain and Phase Measurement of Bandgap Using Stability Analysis

- Stability analysis is a linear small-signal variant of AC analysis.
- Spectre stability analysis performs a Middlebrook stability analysis.
 - In this method, if you connect an ideal **0V** voltage source in series with the feedback path and drive it with a **1V** signal at all frequencies, the loop gain transfer function appears across the voltage source terminals.
- Stability analysis measures the open loop gain and phase in the circuit with the loop closed.
 - The loading in the feedback path is maintained.
 - A single-ended **iprobe** can be instantiated in the schematic to act as the signal source for the stability analysis.

Action 83: From the Library Manager, open the **TB_BGR > tb_bgr > schematic** view.

Action 84: Descend into `BGR_circuit`.

`BGR_circuit` contains an iprobe, `IPRB0`, as shown in Figure 63.

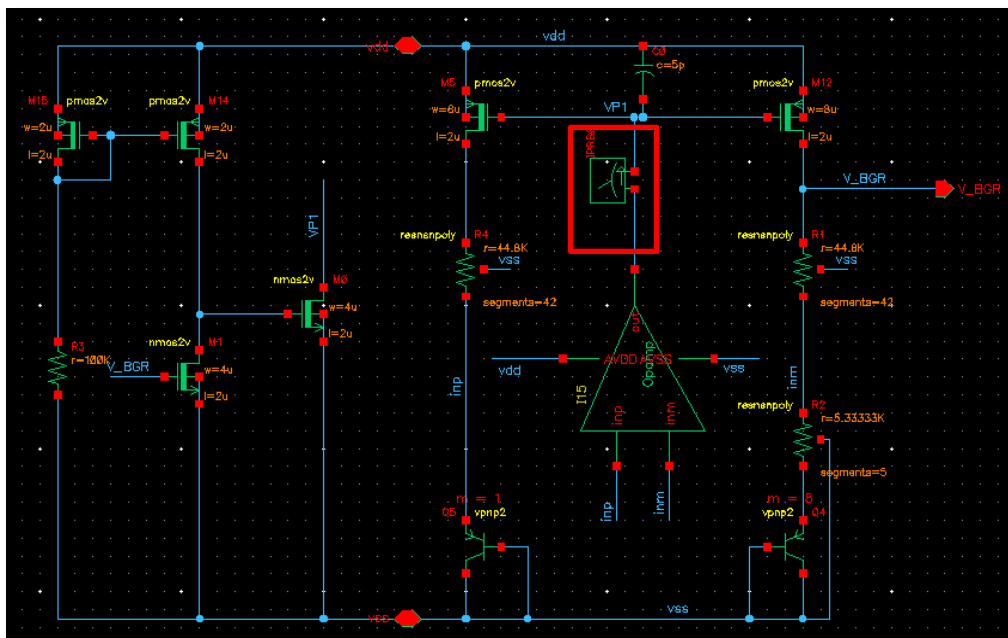


Figure 63: Bandgap circuit Schematic for stability analysis (stb)

Action 85: From the schematic, select **Launch > ADE Explorer**.

Action 86: In the **Launch ADE Explorer** window that opens, select **Create New View** and click **OK**.



Figure 64: Launching ADE Explorer from schematic window

Action 87: In the **Create new ADE Explorer view** window, change the view name to `maestro_Stb`.

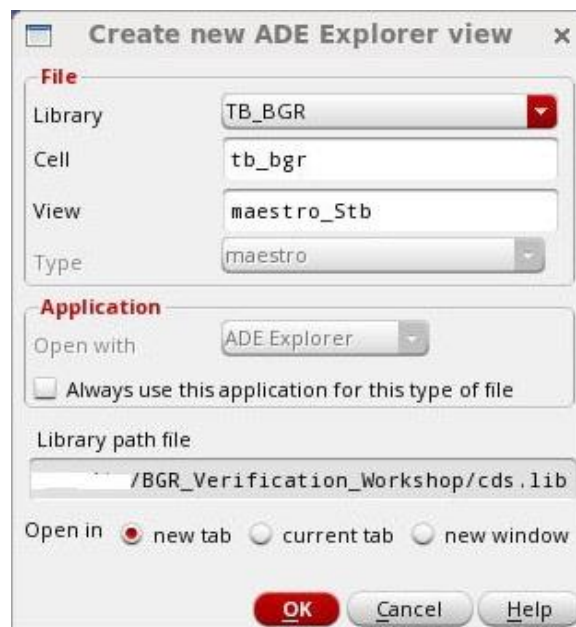


Figure 65: Opening maestro_Stb view of tb_bgr cell

The **ADE Explorer** window appears as shown in Figure 66.

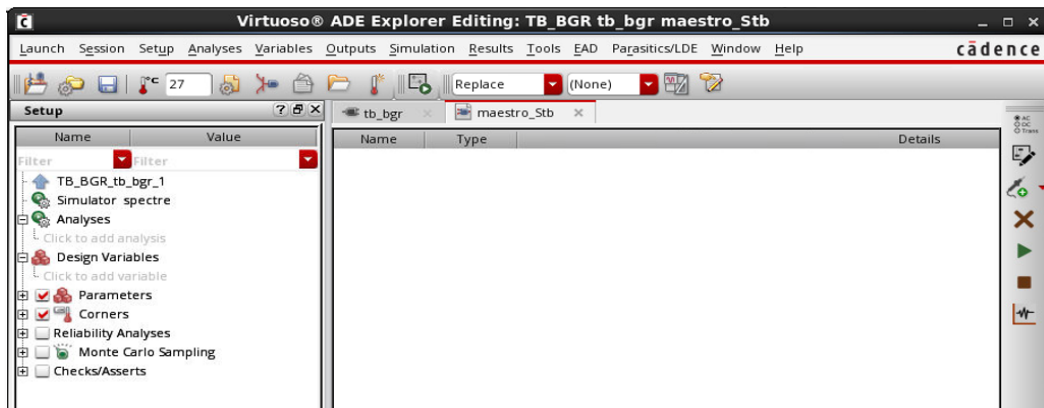


Figure 66: ADE Explorer window showing maestro_Stb view

Action 88: In ADE Explorer, select **Click to add analysis** in the **Analyses** section of the **Setup** assistant.

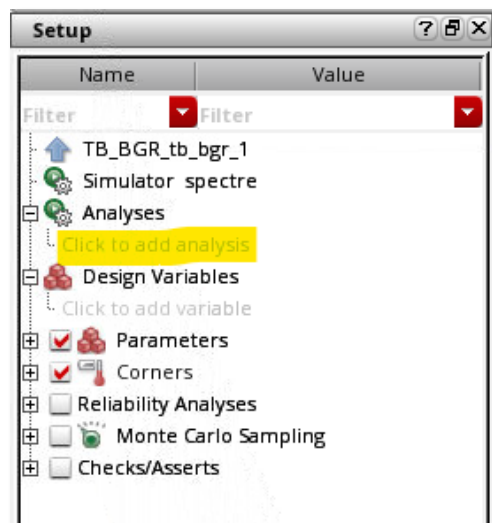


Figure 67: Opening 'Analyses' form from 'Click to add analysis'

Action 89: In the **Choosing Analyses** form, select the **stb** analysis and set up the options as follows:

- Set **Sweep Variable** as **Frequency**.
- Select the **Start** frequency 1Hz and **Stop** frequency for 1GHz.
- Choose **Automatic** for **Sweep Type**.
- For **Probe instance/Terminal**, click on **Select** and descend into **schematic** and select **/I18/IPRB0**.
- For selecting **Local Ground Name**, click on **Select** for the **/vss** net.

Action 90: Click **OK**.

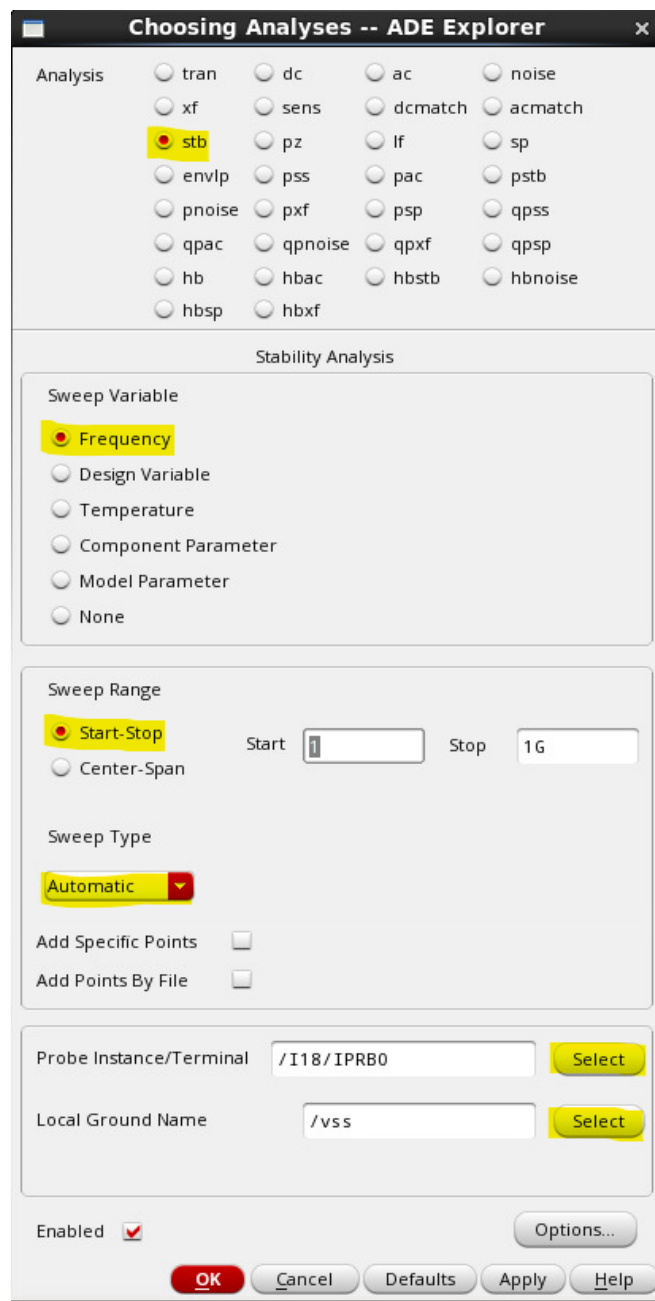


Figure 68: Setup of DC-Sweep Analysis

Action 91: In the **ADE Explorer** window, click on **Variables** and select **Copy From Cellview** as shown in Figure 69.

Action 92: In the **Setup** view assistant, set the **Design Variables** values for **avdd=2** and **avss=0**.

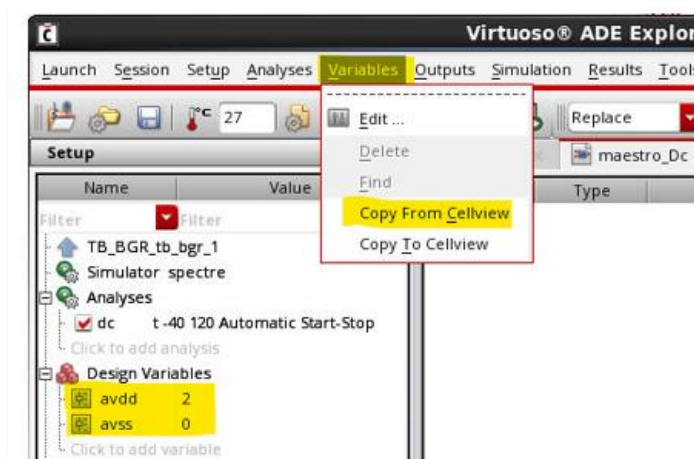


Figure 69: Copying variables from cell view

Action 93: In the **ADE Explorer** window, click on the green button  to **netlist** and **run** the simulation.

Action 94: Select **Results > Direct Plot > Main Form** in the **ADE Explorer** window as shown in Figure 70 to open **Direct Plot Form**.

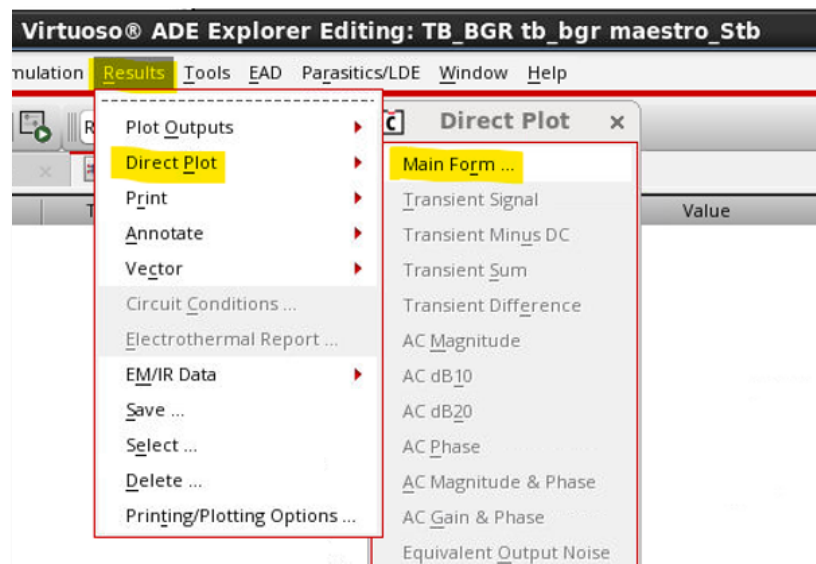


Figure 70: Opening Direct Plot Form from ADE Explorer

Action 95: In the **Direct Plot Form** window, select the following to generate the Bode Plot:

- **Loop Gain** from **Function**
- **Magnitude and Phase** as **Modifier**
- **dB20** as **Magnitude Modifier**
- Enable **Add To Outputs**.
- Click on the **Plot** button.

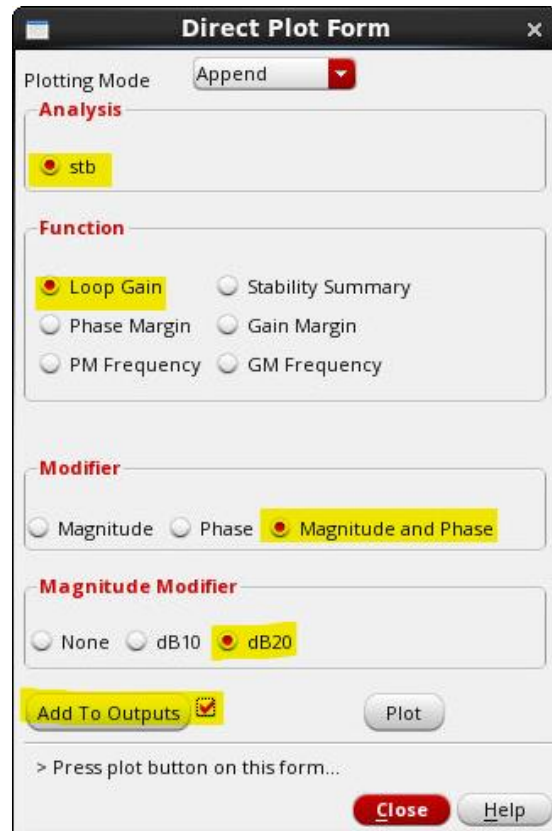


Figure 71: Plotting loop gain from Direct Plot Form

The loop gain waveforms will be plotted as shown in Figure 72.

Action 96: In the **ViVA** window, click **Split Current Strip**  to separate the plots.

Action 97: Add single-point markers using the **M** bindkey on the plot.

Action 98: Place a vertical marker using the **V** bindkey on the plot.

Bandgap Reference Verification: RAK

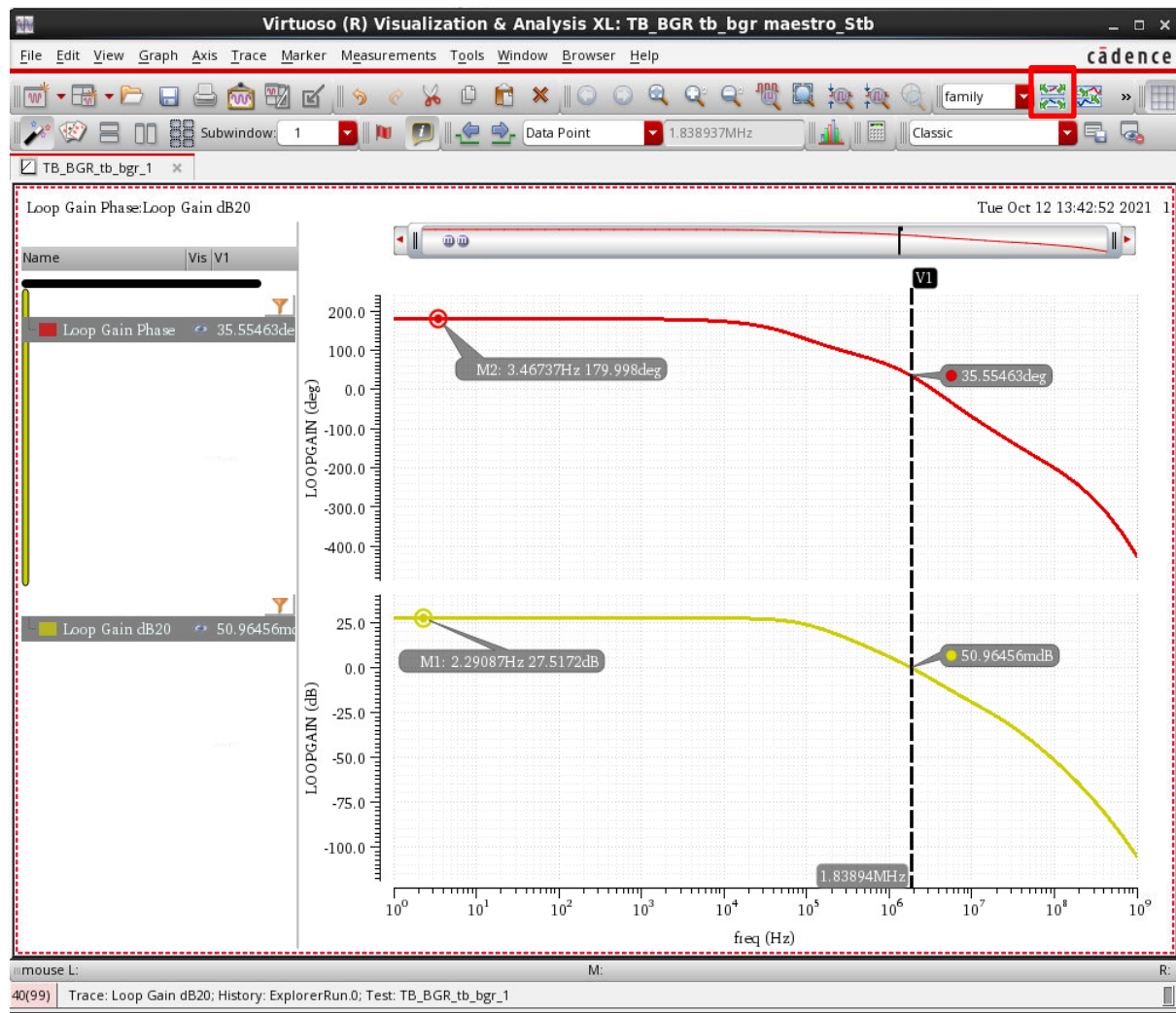


Figure 72: Loop gain and phase plots in VIVA XL window

Action 99: In the **Direct Plot Form** window, select **Phase Margin** as **Function** to show the phase margin value in degrees as shown in Figure 73.

Action 100: Click on **Add To Outputs** to add the phase margin expression to the maestro view.

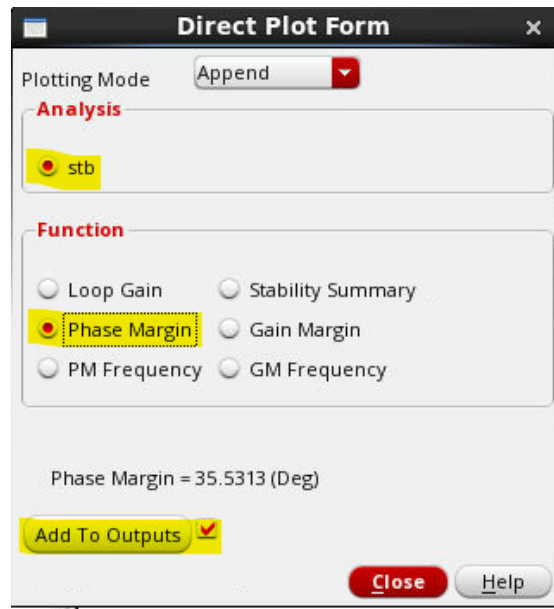


Figure 73: Displayed phase margin in Direct Plot Form

Action 101: In the **Direct Plot Form** window, select **Gain Margin** as **Function** to show the gain margin value in **dB** as shown in Figure 74.

Action 102: Click on **Add To Outputs** to add the gain margin expression to the maestro view.

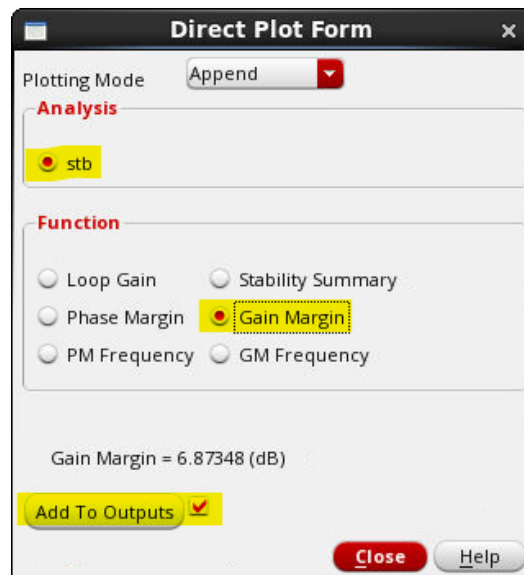


Figure 74: Displayed gain margin in Direct Plot Form

Action 103: In the **Direct Plot Form** window, select **Stability Summary** to show both gain and phase margin values along with frequencies as shown in Figure 75.

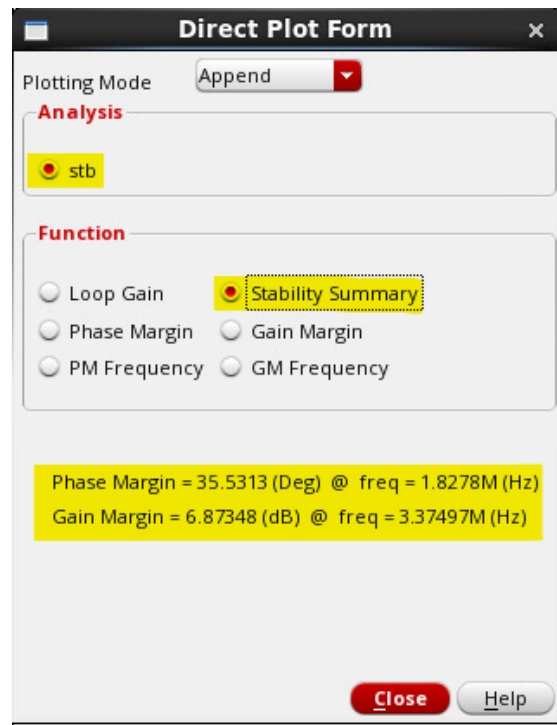


Figure 75: Displayed stability summary in Direct Plot Form

Note: In the **Direct Plot Form** window, you can select **PM Frequency** to get just the phase-margin frequency and select **GM Frequency** to get the gain-margin frequency.

Action 104: The **ADE maestro** view will appear as shown in Figure 76.

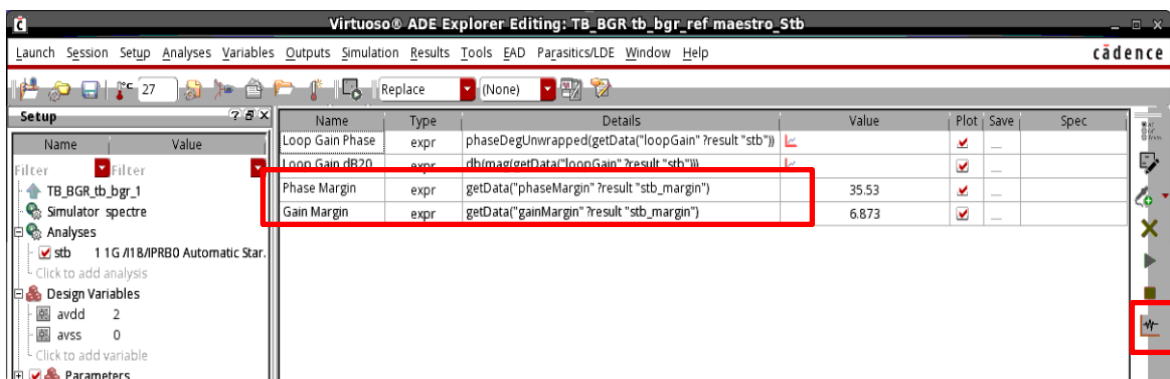


Figure 76: ADE maestro_Stb view

Bandgap Reference Verification: RAK

Action 105: Click **Plot Outputs**

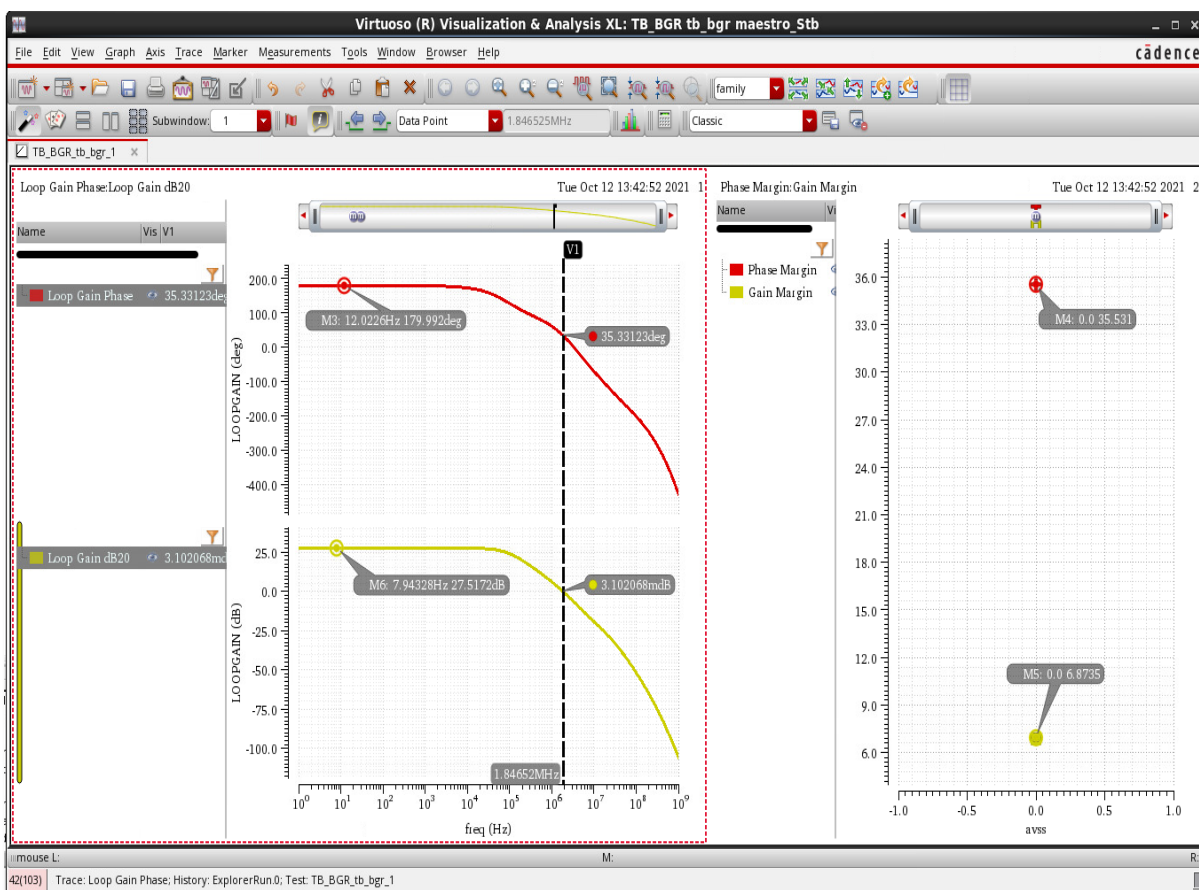


Figure 77: Output plots

Action 106: Close all windows but leave the **Virtuoso** session open for the next lab.

Lab 4: Transient Noise Measurement of Bandgap

- Transient noise calculates the effects of large-signal noise on virtually any system. It determines the impact of noise in the time domain. This is an extension to the current **transient analysis** in **Spectre**.
- This section provides an overview of the transient noise analysis setup form in **ADE**. Later, it provides an overview of **Direct Plot Form** (for plotting transient noise results after the transient simulation).

Action 107: From the **Library Manager**, open the **TB_BGR > tb_bgr > schematic** view. The schematic window opens as shown in Figure 78.

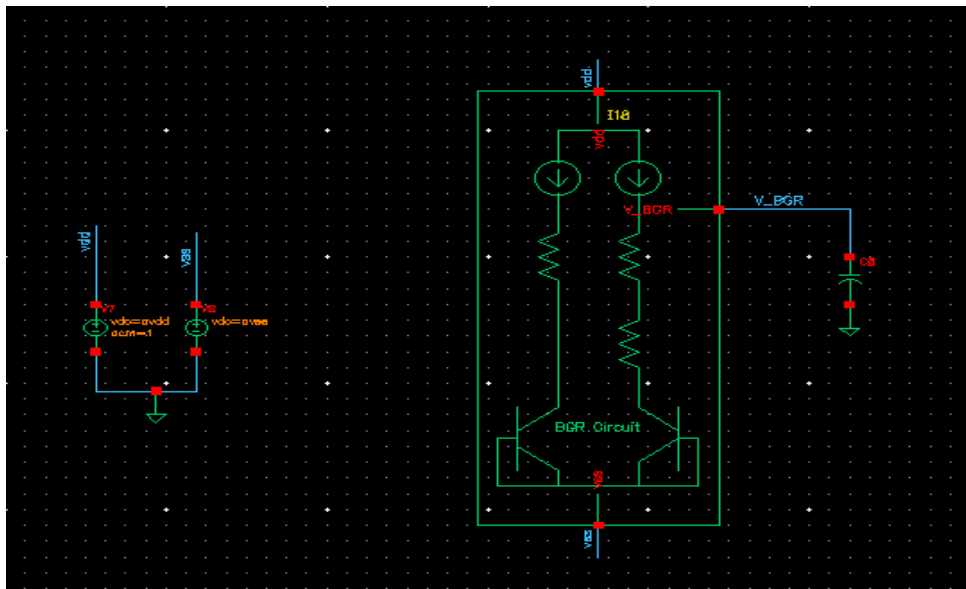


Figure 78: Schematic of tb_bgr cell testbench

Action 108: From the schematic, select **Launch > ADE Explorer**.

Action 109: In the **Launch ADE Explorer** window, select **Create New View** and click **OK**.

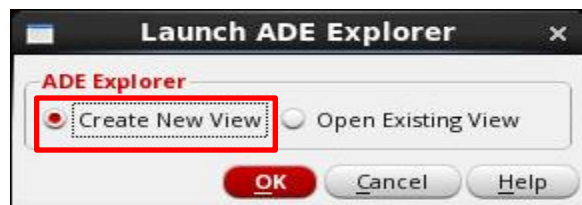


Figure 79: Launching ADE Explorer from the schematic window

Action 110: In the **Create new ADE Explorer view** window, change the view name to `maestro_Trans_Noise` and click **OK**.

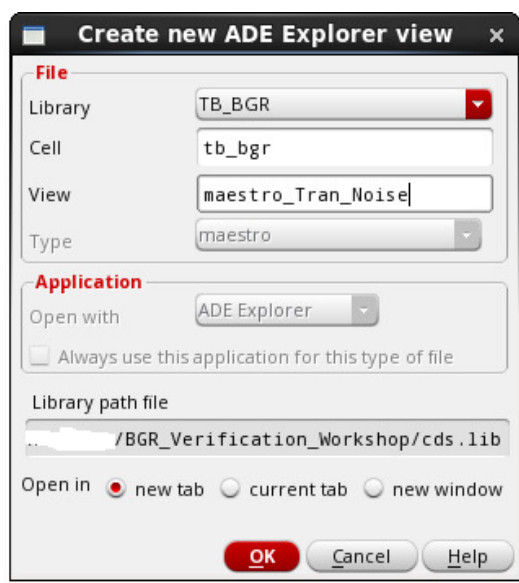


Figure 80: Opening `maestro_Trans_Noise` view of the `tb_bgr` cell

The **ADE Explorer** window opens as shown in Figure 81.

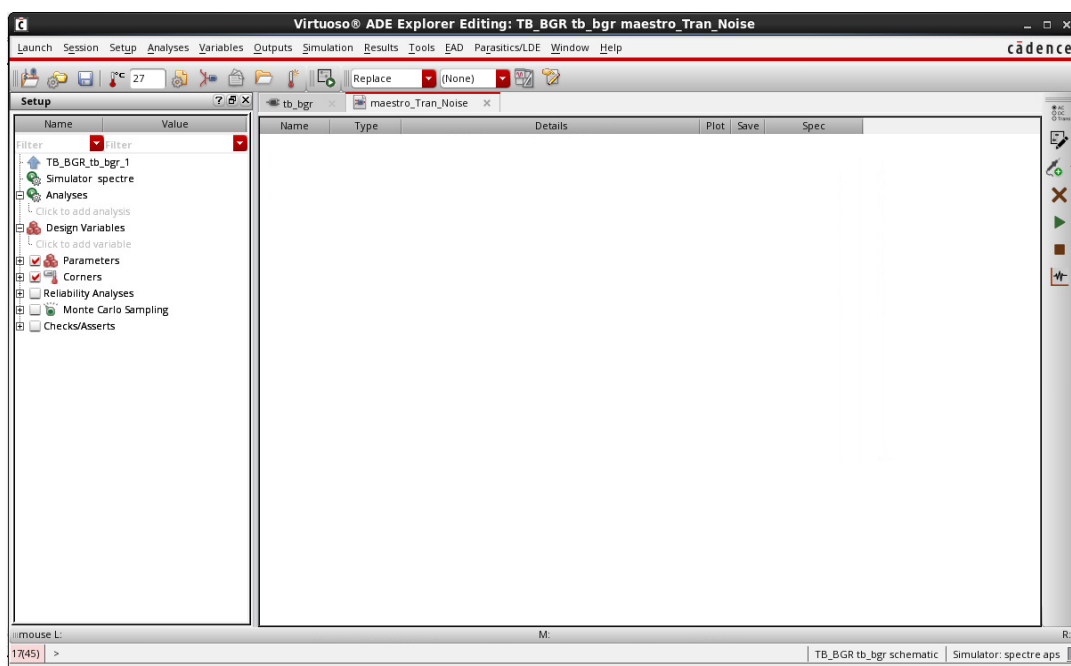


Figure 81: ADE Explorer window showing `maestro_Trans_Noise` view

In **ADE Explorer**, select **Click to add analysis** in the **Analyses** section of the **Setup** assistant as shown in Figure 82.

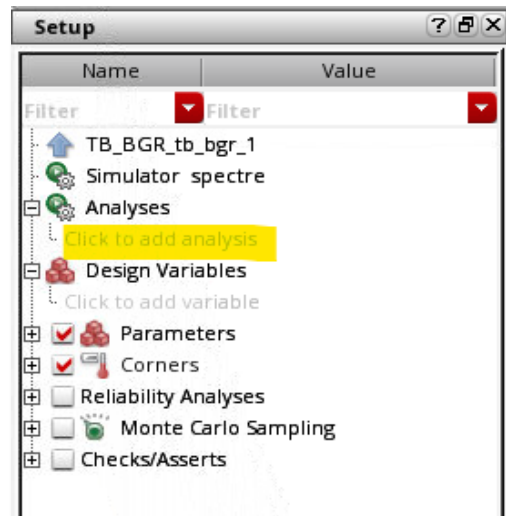


Figure 82: Opening ‘Analyses’ form from ‘Click to add analysis’

Action 111: In the **Choosing Analyses** form, select the **tran** analysis in **Choosing Analyses** form & set the following options as below:

- Enable the **Transient Noise** checkbox.
- Set **Noise Fmax** to **163.84M**.
- Enable the **Fourier Analysis Settings** checkbox.
- Set **Circuit Fundamental Frequency** to **10K**, **PSD Max Frequency** to **100M**, and **PSD Min Frequency** to **10K**.
- Click **OK**.

For more information about setting up transient noise, refer to the article, [Setting up transient noise on a bandgap circuit](#), to get an understanding of parameters values related to transient noise analyses, that is, Noise Fmax, PSD Max Frequency, and so on.

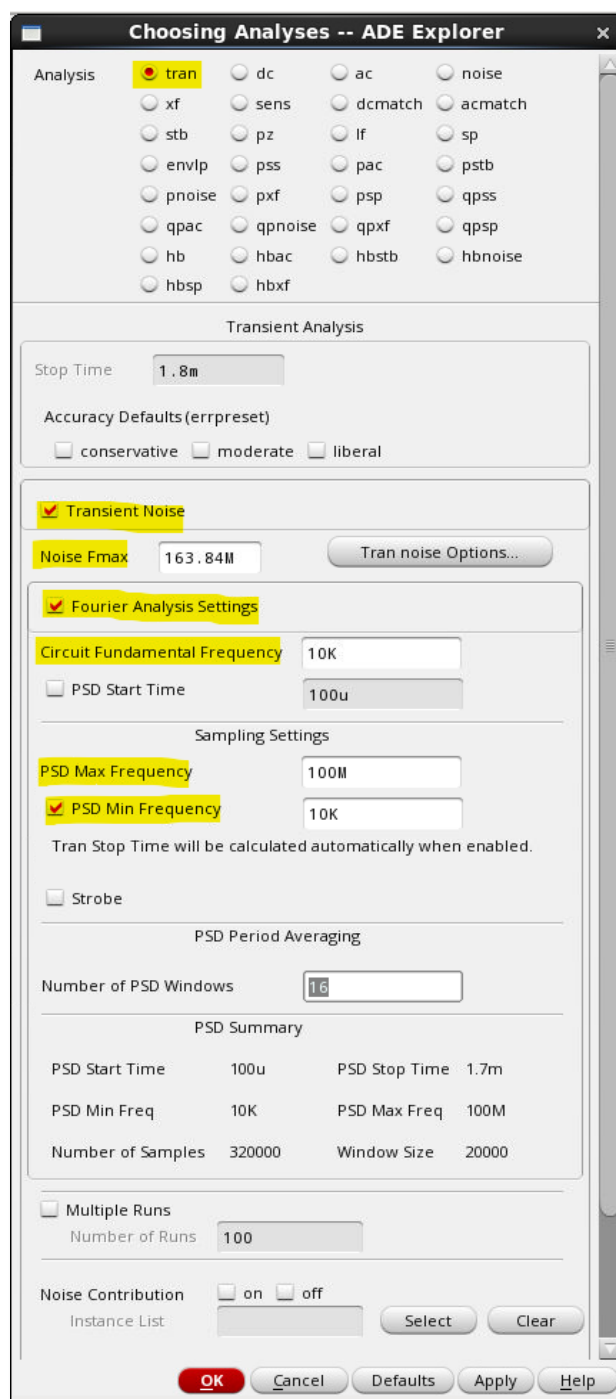


Figure 83: Setup of transient noise analysis

Action 112: In the **ADE Explorer** window, click on **Variables** and select **Copy From Cellview** as shown in Figure 84.

In the **Setup** view assistant, set the **Design Variables** values for **avdd=2** and **avss=0**.

Bandgap Reference Verification: RAK

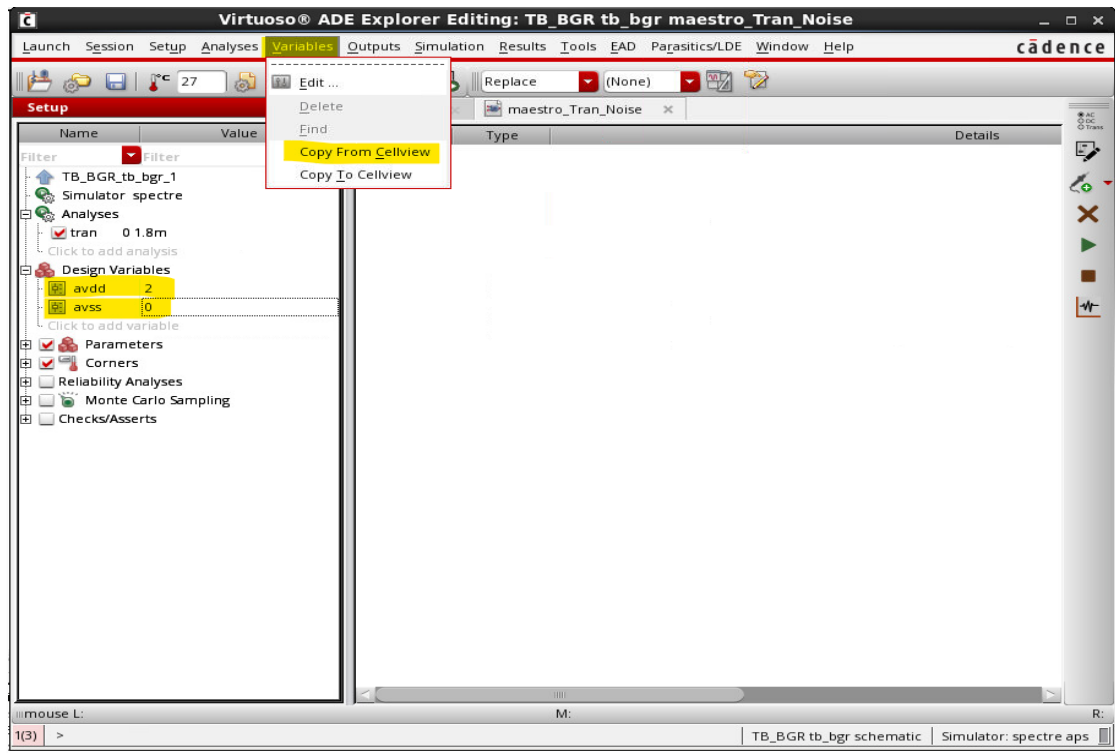


Figure 84: Copying variables from cell view

Action 113: In the **ADE Explorer** window, click on the green button  to **netlist and run** the simulation.

Action 114: Once the simulation finishes, click on **Simulation > Output Log** in the **ADE Explorer** window as shown in Figure 85. Verify that simulation has completed with 0 errors.

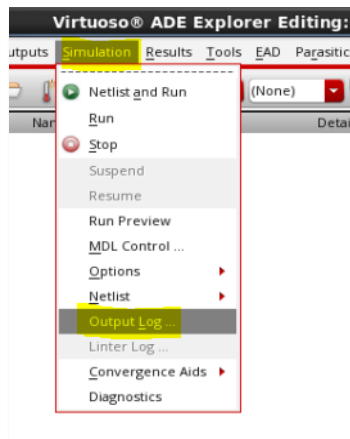


Figure 85: Opening simulation log file from ADE Explorer window

Action 115: Click **Results > Direct Plot > Main Form** in the **ADE Explorer** window as shown in Figure 86 to open the **Direct Plot** form.

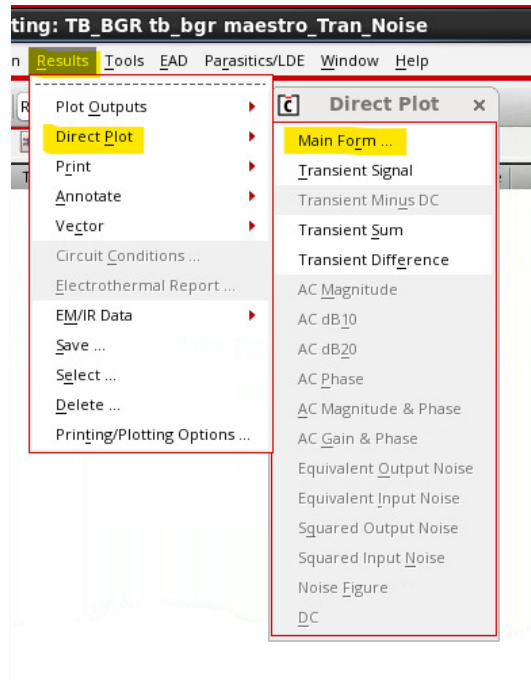


Figure 86: Opening Direct Plot Form from ADE Explorer window

Action 116: In the **Direct Plot Form** window, select **Transient Noise** as **Function**.

Action 117: Select **Net** from the drop-down.

Direct Plot Form will look as shown in Figure 87 with the default PSD settings. Do not change these settings.

Action 118: Click on **Add To Outputs** to add the transient noise expression to the maestro view.

The image shows the 'Direct Plot Form' dialog box in a software application. The 'Plotting Mode' is set to 'Append'. Under the 'Analysis' section, 'tran' is selected. In the 'Function' section, 'Transient Noise' is selected and highlighted in yellow. Below this, 'PSD' is selected over 'PN'. The 'Select' dropdown is set to 'Net'. The 'PSD Default Settings' section shows 'Fundamental Freq' as 10K, 'Bin Number' as 16, and 'Sample per Period' as 20K. The 'Detailed Settings' section includes 'From' (100u), 'To' (1.7m), 'Number of Samples' (320000), 'Window Type' (Rectangular), 'Smoothing Factor' (1), 'Window Size' (20000), 'Detrending Mode' (None), 'Coherent Gain' ((none)), and 'Coherent Gain Factor' (0). At the bottom, there is an 'Add To Outputs' button (highlighted in yellow), 'Default', and 'Replot' buttons. A link '> Select Net on schematic...' is also present, along with 'Close' and 'Help' buttons.

Figure 87: Plotting output noise from Direct Plot Form

Action 119: Select the **IV_BGR** net from the **schematic** window.

Action 120: The Output Noise plot (in dB) will appear as shown in Figure 88.

Action 121: Place single-point markers on the plot using the **M** bindkey.

Bandgap Reference Verification: RAK

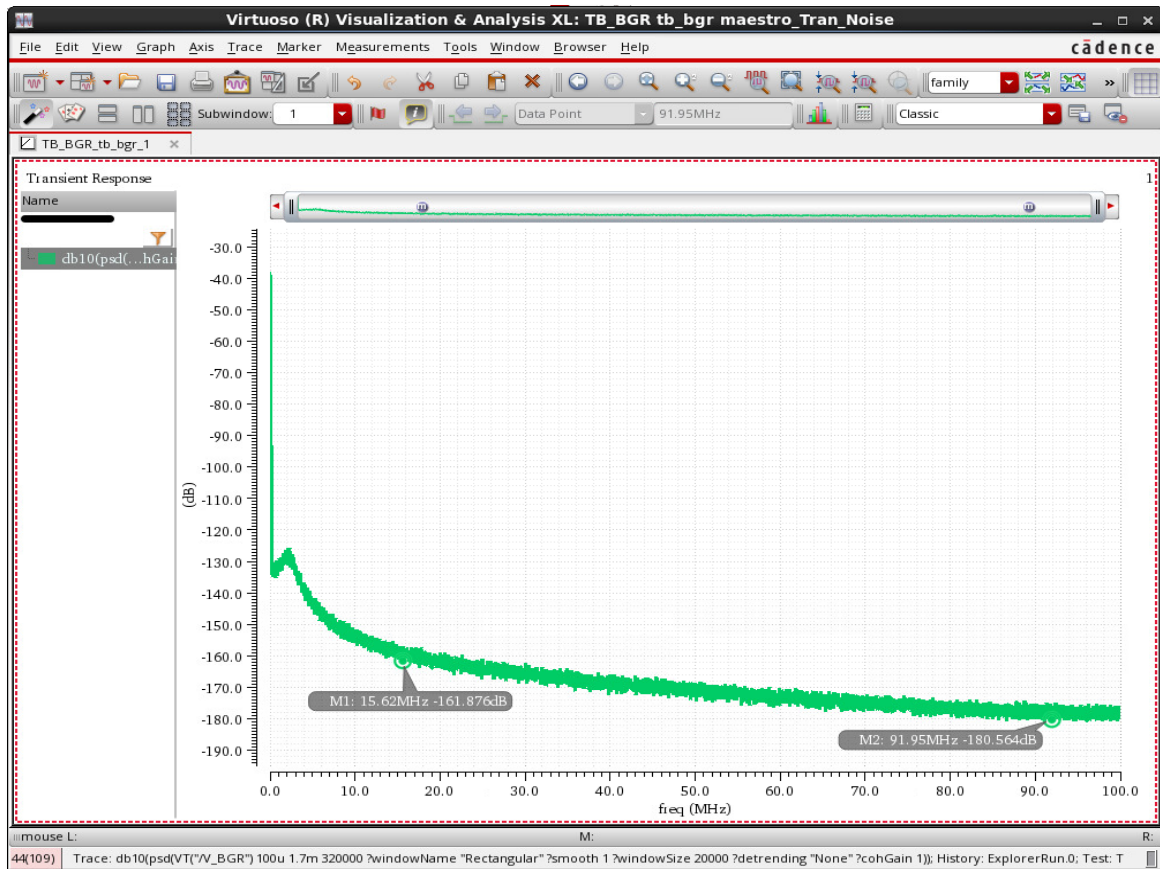


Figure 88: Output transient noise plot in ViVA XL

Action 122: Rename this expression in the maestro view to `Tran_Noise_Plot` as shown in Figure 89.

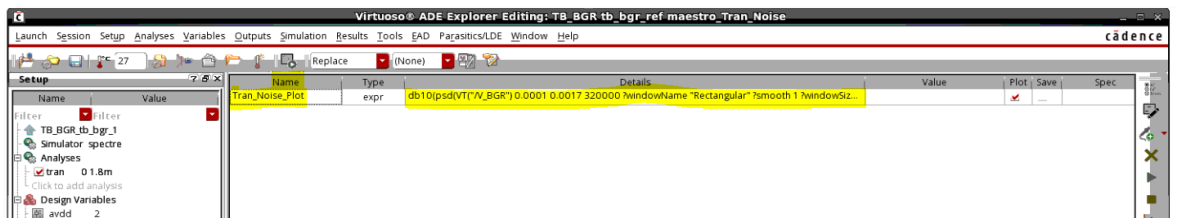


Figure 89: ADE maestro_Tran_Noise view

Action 123: In the **ADE Explorer** window, click **Plot Outputs** .

Action 124: **Close** all windows and the **Virtuoso** session.

This concludes the RAK.

References

1. [Spectre Classic Simulator, Spectre Accelerated Parallel Simulator \(APS\), and Spectre Extensive Partitioning Simulator \(XPS\) User Guide](#)
2. [Virtuoso ADE Explorer User Guide](#)
3. [Virtuoso Visualization and Analysis XL User Guide](#)
4. [Setting up transient noise on a bandgap circuit](#)

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